12. Which of the following statements is true about ethanol?
A. It does not produce toxic gases when it is burnt.
B. It is not produced from sugar cane.
C. It increases accumulation of carbon dioxide in the atmosphere.
D. It is more polluting than conventional fuels.

Explanation: Ethanol:
- ✓ Do not produce toxic gases when burnt.
- ✓ Is produced from sugar cane.
- ✓ Decreases accumulation of carbon dioxide in other atmosphere.

Answer: A

13. The application of biotechnology that involves changing the hereditary material of an organism is:
A. Mechanical engineering
B. Chemical engineering
C. Civil engineering
D. Genetic engineering

Explanation: Genetic engineering is the process of altering an organisms character by inserting another organisms gene is to its DNA.

Answer: D

14. Which of the following is an area of biotechnology application?
Production of:
A. Medicines
B. Computers
C. Soft drinks
D. Plastic bottles

Explanation: Biotechnological applications are practiced in the following areas:
- → Medicine
- → Food
- →Energy
- →Agriculture

Answer: B

15. Production of biogas is an application of biotechnology in the area of:
A. Energy B. Water C. Medicine D. Food

Explanation: Biogas is a flammable mixture of gases that are used as alternative forms of energy.

Answer: A

Post Test – II

Part I: Choose the best answer from the given alternatives.

1. The advantages of genetic engineering over breeding is:
A. It is less expensive than other applications.
B. It is most widely used in developing countries.
C. It is easy to apply by farmers.
D. It always produces a desired trait.

2. The major component of biogas is
   A. Ethanol
   B. Methane
   C. Carbon dioxide
   D. Hydrogen sulfide

3. The first model antibiotic is:
   A. Insulin
   B. Penicillin
   C. Interferon
   D. Hybridomas

4. Fungal proteins produced in biotechnology are:
   A. Antibodies
   B. Mycology
   C. Mycoprotein
   D. Interferon

5. The human hormone produced from genetically engineered bacteria's for diabetic patients is:
   A. Antibody
   B. Insulin
   C. Thyroxin
   D. Adrenaline

6. Blood clotting proteins produced from genetically engineered sheep is sued for people with _____.
   A. Anemia
   B. Leukemia
   C. Goiter
   D. Hemophilia

7. Proteins that make the body less suspiticble to be attacked by viruses are:
   A. Hybridomas
   B. Antibiotics
   C. Insulin
   D. interferon

8. Biogas is produced by the anaerobic effect of:
   A. Protozoa
   B. Bacteria
   C. Viruses
   D. Fungus

9. Biogas is advantageous in all the following except:
   A. It is a renewable source of energy.
   B. It is environment friendly.
   C. It is used for vehicle engines.
   D. It is used for cocking food at home.

10. The production of biogas is an exothermic reaction, because:
    A. It absorbs energy.
    B. It releases energy.
    C. It releases toxics.
    D. It absorbs dust.

11. Which of the following is NOT a raw material used in the production of biogas?
    A. Feaces
    B. Soil
    C. Urine
    D. Plant material

12. The process by which ethanol is separated from other fermentation products is:
    A. Distillation
    B. Crystallization
    C. Decantation
    D. Sedimentation
13. The molasses fermented to produce ethanol is by the action of:
    A. Saccaromycetes
    B. Yeast
    C. Lactobacillus
    D. Streptococcus
14. Enzymes that are used in the production of ethanol are:
    A. Protease
    B. Lipase
    C. Carbohydrase
    D. Carobyxlase
15. Gasohol is the mixture of:
    A. Biogas and fertilizer.
    B. Methane and ethanol.
    C. Ethanol and petrol
    D. Petrol and carbon dioxide.
16. All of the following are the advantages of using ethanol as a fuel, except
    A. It takes a lot of plant material.
    B. It do not produce toxic gases.
    C. It can be mixed and used for vehicle engine.
    D. It has a carbon neutral activity.

Supplementary Exercise on Unit – 7

I. Multiple Choice

1. The use of microorganisms to solve practical problems of humans is
    A. Biometry
    B. Biotechnology
    C. Biodiversity
    D. Biochemistry
2. The advantage of biogas technology is:
    A. The control of plant pests.
    B. The saving of oil and wood.
    C. Source of bitter taste.
    D. Source of carbon dioxide
3. During brewing maltose is used as:
    A. Source of sugar during fermentation.
    B. Culturing medium for the initial growth of yeasts.
    C. Source of bitter taste.
    D. Source of carbon dioxide.

4. The final products of fermentation of sugar by yeast are:
   A. Ethanol and oxygen
   B. Methane and oxygen
   C. Ethanol and carbon dioxide.
   D. Methane and carbon dioxide.
5. The energy that is generated from organic wastes through fermentation is:
   A. Hydropower
   B. Solar energy
   C. Geothermal
   D. Biogas
6. The transfer of genes from the DNA of one organism into the DNA of another organism is known as:
   A. Genetic engineering
   B. Tissue culture
   C. Hybridization
   D. Transplanation
7. Which of the following products of biotechnology is used in medicine?
   A. Tissue culture
   B. Single cell protein
   C. Mycoprotein
   D. Hybridomas
8. Which of the following are the benefits of using mycoprotein as a food?
   A. It has low fat content.
   B. It is commercially available everywhere.
   C. It produces high calorie proteins.
   D. It is the cheapest technology ever.
9. All of the following are the advantages of biotechnology in food production, except
   A. Production of processed food.
   B. Production of wine.
   C. To convert starch in to glucose.
   D. Production of commercial baby food.
10. The first molten antibiotics made from *pencilliumnutatum*. Under which group of microorganisms do we classify this species?
   A. Bacteria
   B. Protozoa
   C. Fungi
   D. Virus
11. Among the applications of biotechnology listed below, which one has brought a relief for diabetic patients?
   A. Discovery of antibiotics.
   B. Tissue culture.
   C. Genetic engineering.
   D. Selective breeding.

12. The first model antibiotic is discovered by:
   A. Joseph Lister
   B. Alexander Fleming
   C. Louis Pasteur
   D. Robert Koch

13. The group of microorganisms that are used in the production of diary products such as cheese and yoghurt are:
   A. Fungi
   B. Bacteria
   C. Viruses
   D. Protozoa

Part II: Matching

Column 'A'	Column 'B'
1. Ethanol	a. Altering organisms genotype.
2. Genetic engineering	b. Component of biogas.
3. Biogas	c. Antibodies produce in culture.
4. Interferon	d. Fugeal protein
5. Hybridomas	e. Proteins that minimize viral attack.
6. Methane	f. Biofuel
7. Mycoprotein	g. Flammable mixture of gases.

Part III. Fill in the blank spaces by using an appropriate.

1. The biological application of microorganisms for human use is _____.
2. The species of bacteria used in lactic fermentation are_____ and _____.
3. The principal species of yeast used in the production of alcoholic drinks is called_____.
4. The action of bacteria's on lactose produces _____.

Answer Key

Post Test I

Part I

1. Yeast
2. Micro organisms
3. Alcoholic Fermentation
4. Anaerobic
5. Honey, Gesho and water

Part III

1. C 2. A 3. B

Post Test – II

Part – I

1. D 2.
6. D 7.
11. B 12

Part I

1. B 2. B
7. D 8. A
13. B

Part II

1. F 2. A

Part III

1. Biotechn
2. Lactoba
3. Saccaro
4. Lactic a

Post Test – II

Part – I

1. D	2.	B	3.	B	4.	C	5.	B	
6. D	7.	D	8.	B	9.	C	10.	B	
11. B	12.	A	13.	B	14.	C	15.	C	16. A

Supplementary Exercise on Unit 7

Part I

1. B	2.	B	3.	A	4.	C	5.	D	6.	A	
7. D	8.	A	9.	B	10.	B	11.	C	12.	B	
13.B											

Part II

1. F	2.	A	3.	G	4.	E	5.	C	6. B 7. D

Part III

1. Biotechnology
2. Lactobacillus and Streptococcus
3. Saccaromycetes
4. Lactic acid

Unit – 8

8. Heredity

Introduction

Organisms share several characteristics with their parents. These characteristics are determined by hereditary information that are transferred during reproduction. The study of heredity is called **genetics**. The life process that is very important in transformation of hereditary characteristics is **cell division**.

8.1 Cell division

Chromosome, Gene and DNA

Chromosome

Chromosomes are thread like structures that are found in the nucleus of a cell. These chromosomes contain the genetic/hereditary information that determines inheritable characteristics of an organism.

> **Inheritable Characteristics** are characteristics that pass from parents to their offspring.

A cell of a given species of an organism contains fixed number of chromosomes in its nucleus. The chromosome number that is found in different groups of organisms will be summarized in the following table.

Organism	Number of chromosome in the nucleus of a cell
Crabs	200
Fern	1200
Human	46
Tomatoes	24
Elephants	56

Sexually produced organism inherits half set of chromosomes from the male parent and the remaining half set from the female parent.

> You inherit half set of chromosomes from your father and the remaining half from your mother.

The chromosomes that come from each parent appears in pair called **homologous pairs**. Accordingly:

→ In human beings each body cell contain 46 chromosomes that appear in 23 homologous pairs.

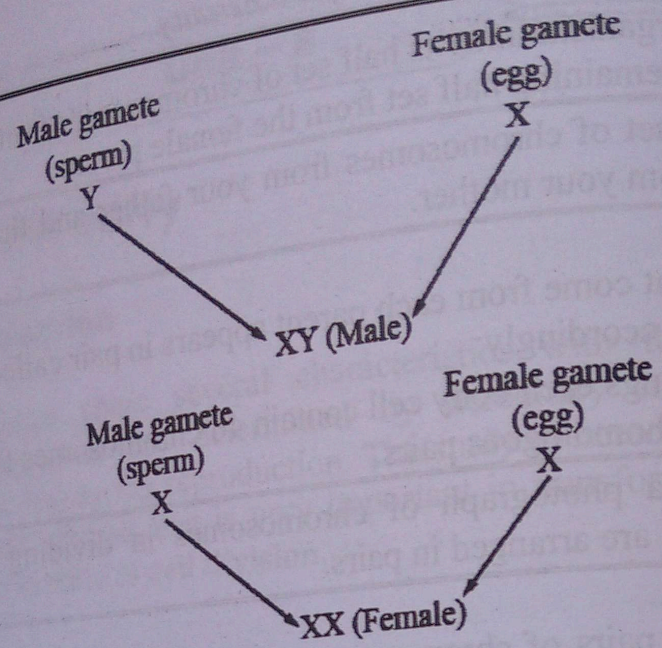
> **Koryotype** – is a photograph of chromosomes in dividing human cells which are arranged in pairs.

Human karyotype 23 pairs of chromosomes in which 22 pairs of the chromosomes are called **autosomes** and the remaining 1 pair is called **sex chromosomes**.

- ✓ **Autosomes** – are chromosomes other than sex chromosomes. The contain genes that determine several characteristics other than determination of sex.
- ✓ **Sex chromosomes** – are chromosomes that determine the sex of an individual. Weather it's male or female. These are **x- chromosome** and **y- chromosome**.
  - In human x – chromosomes is larger than y – chromosomes. This implies that x – chromosomes contain more genes then y – chromosomes.
  - Y – Chromosomes contain genes only for maleness.
  - Human males' has **xy chromosomes**. This means, they inherit **x** – chromosomes from their mother and y – chromosomes from their father.
  - Females have xx – chromosomes which they inherit x – chromosomes from both parents.

> **Note:** 50% of the sperm cells produced contain x – chromosomes and the remaining 50% contain y – chromosomes.

- Chromosomes are made up of a protein coat **histone** and **DNA**.



The probability of having a boy or a girl is proportional. It is 50%.

Based on the number of chromosomes they contain human cells can be divided in to two types. These are:

- ✓ **Somatic cells** – are cells that contain 46 – chromosomes in 23 pairs. Such cells are called**diploid**. All the body cells except sperm and egg cell are somatic cells.
- ✓ **Sex cells** – contain half the chromosome number of those contained in somatic cells. In humans somatic cells contain 23 - chromosomes. They are called**haploid**.
  Examples are sperm cell and egg cell.

Critical thinking
Why do you think sperm cells and egg cells contain half set of chromosomes unlike the somatic cells?

Genes

Genes are small segments of a chromosome. They are also known as units of heredity.
Genes are:

- ✓ responsible for determining specific characteristics of an organism.
- ✓ located on a chromosome like beads on necklace.

A segment or part of a gene is called **allele**. Alleles are the alternative forms of a gene. A gene has two alleles coding for similar or different

traits. The position th
loci.

DNA (Dexoyrib

DNA is an amazing
chromosome.
DNA was introduc
biologists **Maurice**
Francis Crick. T
contribution in a sc
scale. For thisachiev

Critical Thinki

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such as books a
biologists who h
have found with

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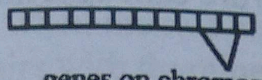
- Phosphate
- Nitrogenc
  up a nucle
  - A
  - T
  - C
  - G

The nitrogenous b
the first letter of t
Among the four n

traits. The position that a gene occupies on a chromosome is known as **loci**.



genes on chromosome

DNA (Dexoyribonucleic acid)

DNA is an amazing molecule that makes up the back bone of a chromosome.
DNA was introduced for the first time in 1950's by the team of biologists **Maurice Wilkins, Rosiland Frnaklin ,James Watson** and **Francis Crick**. These team of biologists have made a massive contribution in a science of heredity and the study of biology at large scale. For thisachievement they have won a Nobel prize award.

Critical Thinking

Have you ever heard about Nobel prize? Refer to different sources such as books and internet. Then list out the name of some of the biologists who here won the Nobel prize award and discuss what you have found with your classmates.

What is DNA?

DNA is a biological molecule which is constructed from smaller molecules known as **nucleotides**.
Anucleotide is composed of the following major components. These are:

- Deoxyribase pentose sugar
- Phosphate unit
- Nitrogenous bases – there are four nitrogenous bases that makes up a nucleotide. These are:
    - Adenine (A)
    - Thymine (T)
    - Guanine (G)
    - Cytosine (C)

The nitrogenous bases that make up a DNA molecule are represented by the first letter of their name.
Among the four nitrogenous bases:

- ✓ Cytosine and Thymine are called **purines.**
- ✓ Adenine and Guanine are called **pyrimidines.**

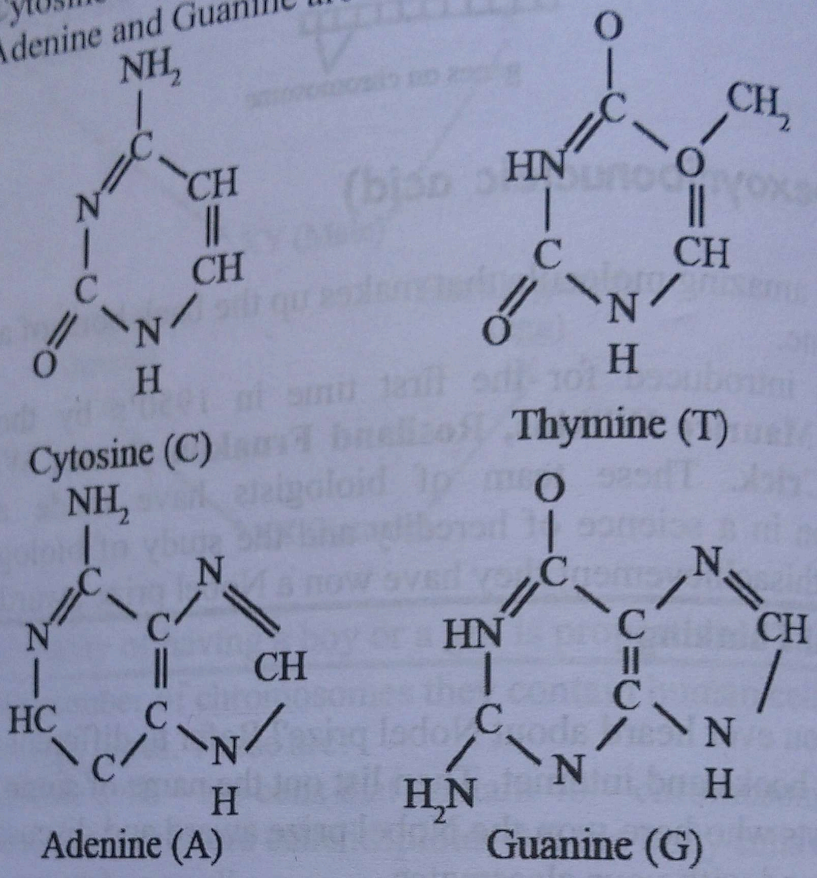


Figure 8.1: Nitrogenous bases

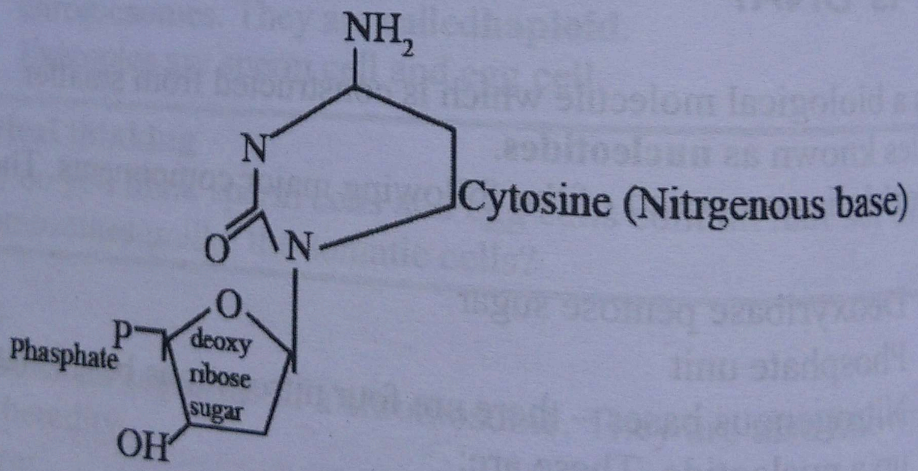


Figure 8.2: A nucleotide

Note: The four nucleotides join in different patterns to make a *DNA* molecule.

A DNA molecule is made up of two (double) chains of nucleotides that are joined to one another through the nitrogenous bases found on each nucleotide. Accordingly, when the two chains join:

- ✓ Adenine (A) pairs with thymine (T)
- ✓ Guanine (G) pairs with cytosine (C)

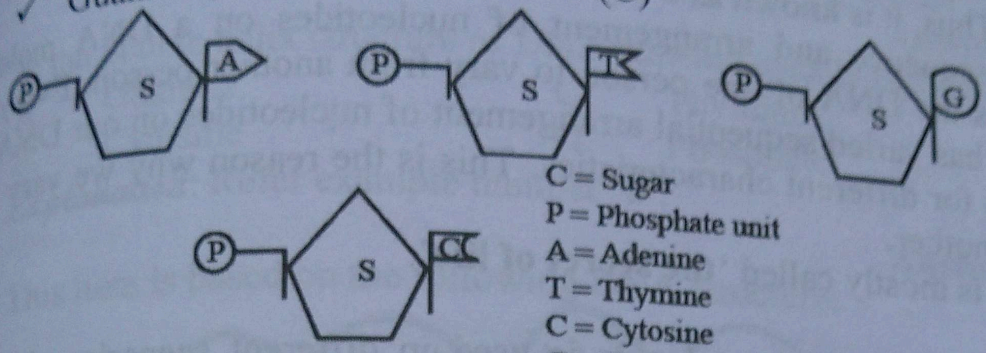


Figure 8.3: Nucleotides bases

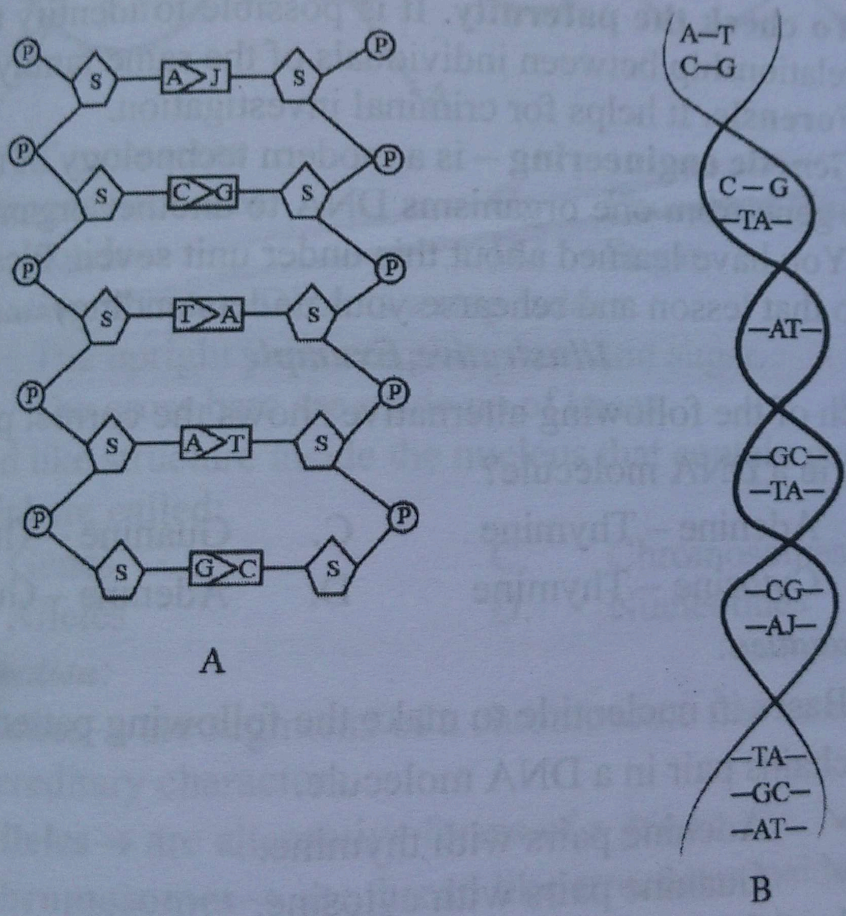


Figure 8.4: A DNA molecules

A model structure of DNA molecules on figure 'A' seems a step ladder, so to simply understand the structure of DNA molecule lets compare it with a step ladder.

- ✓ The upright part is made up of phosphate and sugar.
- ✓ The cross bars are made up of nitrogenous bases.

A DNA molecule is not as short as shown on the above figure. It is composed of a long chain of nucleotide that contains hundreds to thousands of nucleotide chains. Thus it is called **polynucleotide chain**. Actually the structure of DNA molecule is twisted as shown on Figure 'B'. Thus, it is known as a **double helix** molecule.

The sequence and arrangement of nucleotides on a DNA molecule makes the DNA of one person to vary from another person. Every one of us has varied sequential arrangement of nucleotides on our DNA that codes for different characteristics. This is the reason why we vary from one another.

DNA is mostly called '**the secret of life**'.

Now a day's DNA molecule is used in different scenarios. Some of these are:

- ✓ **To check the paternity**. It is possible to identify the genetic relationship between individuals of the same family.
- ✓ **Forensic**. It helps for criminal investigation.
- ✓ **Genetic engineering** – is a modern technology of transferring a gene from one organisms DNA to another organisms DNA. (You have learned about this under unit seven. Please go back to that lesson and rehearse your understanding).

Illustrative Example

1. Which of the following alternative shows the correct pairing of bases in a DNA molecule?

   A. Adenine – Thymine C. Guanine – Thymine
   B. Cytosine – Thymine D. Adenine – Guanine

   Explanation:

   - Bases in nucleotide to make the following pattern nucleotide chains pair in a DNA molecule.
     - ✓ Adenine pairs with thymine.
     - ✓ Guanine pairs with cytosine.

   Answer: A

2. Which of the following fertilization of sex chromosomes will have a boy offspring?

   A. Y chromosome of mother with X chromosomes of father.
   B. Y chromosome of mother with Y chromosome of father.
   C. X chromosome of mother with Y chromosome of father.
   D. X chromosome of mother with X chromosome of father.

Explanation:
- ✓ A boy inherits X – chromosome from his mother and Y – chromosome from his father.
- ✓ A girl inherits X – chromosome from both parents.

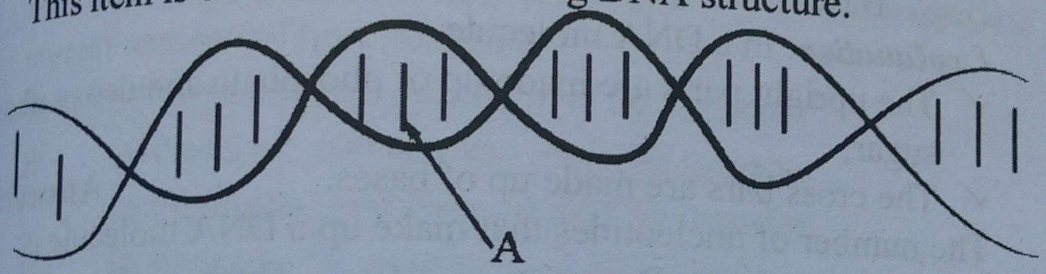
Answer: C

3. In the double helix structure of DNA adenine usually pairs with
   A. Guanine C. Thymine
   B. Cytosine D. Phosphate

Explanation: Refer example number '2'.

Answer: C

4. This item is based on the following DNA structure.



Letter "A" represents
A. Nitrogenous bases C. Phosphate group
B. Nucleotide D. Sugar

Explanation: Assume DNA is a step ladder:
- ✓ The upright parts are phosphate and sugar.
- ✓ The cross bars are made up of bases.

Answer: A

5. Thread like structure inside the nucleus that contain genetic material are called:
   A. Genes C. Chromosomes
   B. Alleles D. Nucleotides

Explanation:
- **Genes**→ are segments of a chromosome that contain hereditary characters.
- **Alleles**→ are alternative forms of a gene.
- **Chromosomes**→ are thread like structures inside the nucleus and contain genetic materials.
- **Nucleotides**→ are molecules, that make up a DNA molecule.

Answer: C

6. Which of the following includes all the rest?
   A. Phosphate C. Deoxyribose
   B. Adenine D. Nucleotide

Explanation: A nucleotide consists of phosphate unit, deoxyribose and nitrogenous bases (Adenine, Thymine, Cytosine and Guanine).

Answer:

7. RNA is different from DNA, because RNA contains uracil base instead of:
   A. Thymine B. Cytosine C. Adenine D. Guanine

Answer:

8. If one imagines DNA molecule as a step ladder, the cross bars of the ladder are made up of:
   A. Sugar and phosphate C. Sugar
   B. Bases D. Phosphate unit

Explanation: In a DNA molecule
- ✓ The upright parts are made up of phosphate and deoxy ribose sugar;
- ✓ The cross bars are made up of bases.

Answer:

9. The number of nucleotides that make up a DNA molecule is:
   A. One B. two C. Three D. Many

Explanation: DNA is a big molecule that is made of a long chain of nucleotides.

Answer:

10. If one of the strand of the two chains of nucleotides has a bases ACGT bases, the corresponding bases on the other chain are:
    A. TGCA B. AGCA C. TGCT D. TCGT

Explanation: In a DNA molecule Adenine base always pairs with thymine and cytosine pairs with guanine. Thus the corresponding pair of bases on the other pair of nucleotide chain is TGCA.

Answer:

11. If a DNA molecule has 1400 bases from which 400 are adenine bases. What will be the number of thymine, guanine and cytosine bases respectively?
    A. 300 thymine, 400 guanine and 300 cytosine.
    B. 300 thymine, 300 guanine and 400 cytosine.
    C. 400 thymine, 300 guanine and 400 cytosine.
    D. 300 thymine, 350 guanine and 350 cytosine.

Explanation: In a DNA molecule:
- ✓ The number of Adenine is equal to the number of Thymine;
- ✓ The number of Cytosine is equal to the number of Guanine.

Answer:

12. When DNA molecule replicates, the daughter DNA molecules will each consist of:
    A. Two new chains of nucleotides
    B. Two old chains of nucleotides
    C. Two old and two new chain of nucleotides
    D. One old and one new chain of nucleotides

Explanation: A DNA molecule is made of tow strands of nucleotides. During DNA replication, one strand of nucleotides is new and the other strand is old. **Answer: D**

13. If a DNA molecule has 100 cytosine and 50 thymine bases, the correct number of base pairs are:
    A. 100 thymine, 50 guanine and 50 adenine
    B. 50 thymine, 100 guanine and 100 cytosine
    C. 100 thymine, 100 guanine and 50 cytosine
    D. 50 thymine, 50 guanine and 100 guanine

Answer: B

14. Which of the following nitrogenous bases of DNA are equal in number?
    A. Cytosine and guanine
    B. Thymine and cytosine
    C. Adenine and cytosine
    D. Adenine and guanine

Explanation: In a DNA molecule the number of:
- ✓ Adenine equals to thymine;
- ✓ Cytosine equals to guanine.

Answer: A

15. Which of the following alternatives shows the correct pairing of bases in a DNA molecule?
    A. Adenine- Guanine
    B. Guanine- Thymine
    C. Cytosine- Thymine
    D. Adenine- Thymine

Explanation: A DNA molecule is made up of four types of nitrogenous bases. These are Adenine (A), Thymine (T), Cytosine (C) and Guanine (G). Thus:
- ✓ Adenine pairs with thymine
- ✓ Cytosine pairs with guanine

Answer: D

Quick Check

1. List out the major components of a nucleotide.
2. While saying a DNA is a polynucleotide molecule, what does it mean to say?
3. If the sequence of bases on one strand of a DNA molecule is as follows; what do you think will be the bases pairs on the other stand?

   ATCGGCTAATCAGT
4. There is a saying that 'DNA is a secrete of life.Can you explain reason why this is said?
5. If the number of bases on a DNA molecule is 1600 and among these, the number of Thymine base is 500. What do you think are the numbers of adenine, guanine and cytosine bases respectively?

Post Test – I

I. Choose the best answer from the given alternatives.

1. Which one of the following is NOT a component of a nucleotide?
   A. Deoxyribose sugar
   B. Phosphate unit
   C. Nitrogenous bases
   D. Ribose Sugar
2. The correct pairing of nitrogenous bases in DNA molecule is:
   A. Adenine – Cytosine
   B. Guanine – Thymine
   C. Adenine – Thymine
   D. Cytosine – Thymine
3. Among the following bases, which one is NOT found in a DNA molecule?
   A. Adenine
   B. Uracil
   C. Cytosine
   D. Guanine
4. In one of the strands of nucleotides that makes up a DNA molecule, it has bases ACGTAGT. Thus, what will be the bases sequence on the corresponding nucleotide chain?
   A. GCTACGT
   B. TGCATCA
   C. GCACGTA
   D. TGGATCA
5. In a DNA molecule having 2400 bases; if the number of cytosine base is 750, what will be the number of adenine, thymine and guanine bases respectively?
   A. 750, 750, 250
   B. 450, 750, 450
   C. 450, 450, 750
   D. 750, 450, 750

6. In a DN
   A. A
   B. A
   C. A
   D. A
   The nu
7. A. 2
   Which
8. A.
   In hum
9. A.
10. A DN
    Guani
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    A.

Cell Cycle

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out
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DNA re
copy.

ii) M

Mitosis is t
major stage
✓ Pr
✓ M
✓ An
✓ Te

6. In a DNA molecule, which of the following is necessarily true?
   A. Adenine = Cytosine
   B. Adenine + Thymine = Cytosine + Guanine
   C. Adenine = Thymine + Guanine + Cytosine
   D. Adenine + cytosine = Thymine + Guanine
7. The number of chromosomes in human somatic cells are;
   A. 23 B. 46 C. 72 D. 92
8. Which of the following includes all the rest?
   A. Gene B. Nucleotide C. Chromosome D. Allele
9. In humans, which sex chromosomal combinations result in male?
   A. XX B. XY C. XYX D. XXY
10. A DNA molecule has the following bases; Adenine = 400, Guanine = 600, what is the number of bases found in this DNA molecule?
    A. 800 B. 1800 C. 2000 D. 1400

Cell Cycle

Cell cycle is sequence of events which occurs between one cell division and the next division. The process of cell cycle has three main stages:

i) Interphase

Interphase is a period of synthesis and growth. It is:

- ✓ the longest phase of the cell cycle.
- ✓ also known as the non – division stage.
- ✓ a cell produces materials required for its growth and carrying out all its functions.
- ✓ Stage where DNA replication occurs.

DNA replication is the process by which a DNA makes it's own copy.

ii) Mitosis

Mitosis is the stage that involves mainly the nuclear division. It has four major stages. These are:

- ✓ Prophase
- ✓ Metaphase
- ✓ Anaphase
- ✓ Telophase

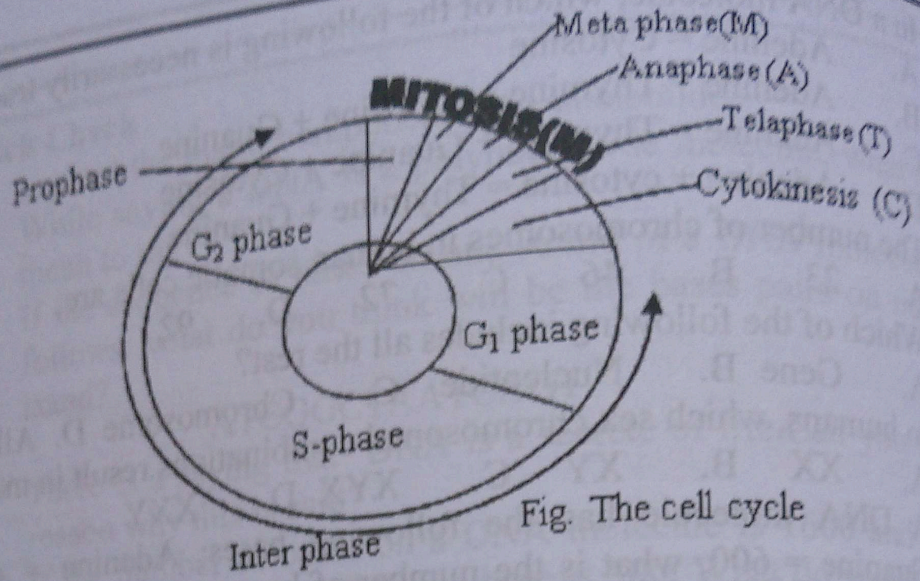


Fig. The cell cycle

Figure 8.5: Cell cycle

iii) Cytokinesis

Cytokinesis involves the division of cytoplasm into two daughter cells.

Phase	Events within a cell
G1 (Gap 1)	Intensive cellular synthesis including new cell organelles. Cell growth occurs.
S (synthesis)	DNA replication occurs
G2 (Gap 2)	Intensive cellular synthesis, mitochondria and Chloroplast divide. Mitotic spindle begins to form.
M (Mitosis)	Nuclear division occurs in four phases.
C (Cytokinesis)	Equal distribution of organelles and cytoplasm in to each daughter cells.

Meiosis and Mitosis

There are two types of cell division. Namely; meiosis and mitosis.

i) Mitosis

- Mitosis is a kind of cell division in which daughter cells produced have identical and equal number of chromosomes with the parent cell.
- The daughter cells produced contain the same genetic information as that of the parent cell.

- The process of mitosis has the following basic features:
  - ✓ Daughter cells and parent cells have equal number and identical type of chromosomes.
  - ✓ Takes place during the division of somatic cells to produce other somatic cells.
  - ✓ It takes place during the life processes such as:
    - Growth
    - Replacement of worn out and damaged cells.
    - Asexual reproduction in unicellular organisms.

Somatic cells are all the cells with exception of sex cells (gametes).

- ✓ Two daughter cells are produced from a single parent cells.
- ✓ Diploid cells are produced.

Diploid (2n) is a cell that contains full set (homologous pair) of chromosomes.

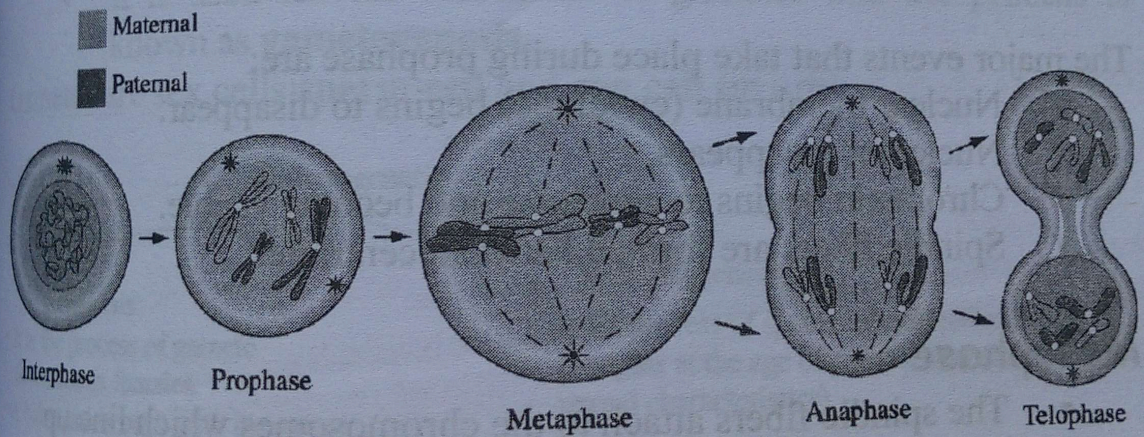


Figure 8.6: Mitosis

The process of mitosis occurs in four major stages. These are:

- ✓ Prophase
- ✓ Metaphase
- ✓ Anaphase
- ✓ Telophase

The events that occur in the nucleus during mitosis are usually observed in cells which have been fixed and stained.

Interphase

During interphase, the following events will take place:

- ✓ The DNA replication and chromosomes are copied to make identical chromosome pairs called **sister chromatids**.
- ✓ A cell carries out its normal activities.

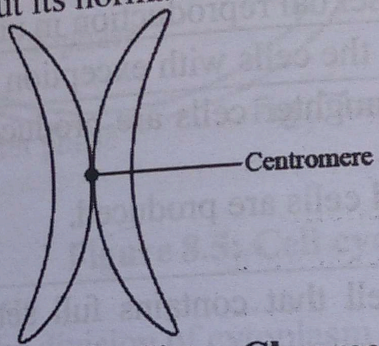


Figure 8.7: Sister Chromatids

Prophase

The major events that take place during prophase are:

- ✓ Nuclear membrane (envelope) begins to disappear.
- ✓ Nucleolus disappears.
- ✓ Chromatin begins to condense and become visible.
- ✓ Spindle fibers are formed between centrioles.

Metaphase

- The spindle fibers attach to the chromosomes which line up along the cell's mid plane.
- The chromosomes align at the equator.

Anaphase

- Chromatids separate at centromeres, and one group of chromosome moves toward each pole.
- In other words, sister chromatids separate at centromere and each chromosome moves towards the opposite pole.

Telophase

- Chromosomes arrive at each pole.

- Chromosomes become long and thin losing ability to be seen clearly.
- Nucleolus reappears.

Cytokinesis

Cytokinesis takes place and results in forming two daughter cells.

Cytokinesis is a division of cytoplasm.

ii) Meiosis

Meiosis is a cell division in which the chromosome number of the parent cell is reduced by half in daughter cells. Thus, meiosis is also known as **reduction division.**

The major features of Meiosis are:

- ✓ The parent cell and daughter cells are not identical.
- ✓ Haploid (n) cells are formed.
- ✓ Four daughter cells are produced from a single parent cell.
- ✓ It is used for the formation of gametes and the process is known as **gametogenesis.**

Gametes are sex cells;**sperm cell** and **egg cell (ovum)**.

Gametogenesis

Oogenesis

- It is the process of gamete formation is females.
- A baby girl is born with immature ova (egg) in the ovary.
- Oogenesis in girls starts before birth.
- When the girls reach the age of puberty, the immute eggs in the ovary starts to mature in a monthly cycle.

Spermatogenesis

- Is the process of sperm cell formation in males. It begins at the age of puberty (the age of 2ndary sexual characteristics)

Puberty is the stage at which boys and girls start to experience secondary sexual characteristics. (You will learn the detail under unit 9).

The gametes that are produced by meiosis are not exactly like. This occurs because of the combination of chromosomes is different and exchange of genes between chromosomes occur by the processes known as **synapsis** and **crossing over**.

Synapsis and Crossing over

Synapsis is the process in which homologous chromosomes come length wise side by side. During these times they exchange genetic material (DNA) with one another by the process called **crossing over**. Crossing over produces new combination of genes. This greatly enhances genetic variation.

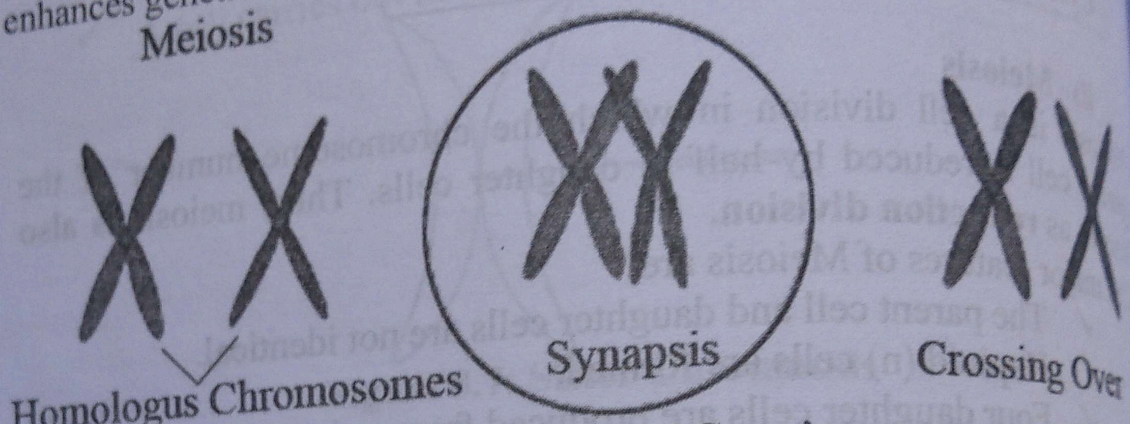


Figure 8.8: Synapsis and Crossing over

Meiosis I and Meiosis II

The process of meiosis occurs in two major stages. These are

- ✓ **Meiosis– I** involves separation of homologous chromosomes.
- ✓ **Meiosis – II** involves separation of sister chromatids.

i) **Meiosis – I**

Meiosis – I is when homologous chromosomes separate. It occurs through four stages. These are:

- ✓ Prophase – I
- ✓ Metaphase – I
- ✓ Anaphase – I
- ✓ Telophase – I

Before meiosis starts the interphase (replication of DNA) occurs.

Prophase – I Chromosomes appears in condensed form.

- Nuclear envelope (membrane) disappears.
- Homologous chromosomes synapse. In other words homologous chromosomes associate with one another.

Metaphase – I

- ✓ Spindle forms
- ✓ Pairs of chromosome line up on the metaphase plate.
- ✓ Crossing over (exchange of genetic material between homologous chromosomes) occur

Anaphase – I

- ✓ Homologous chromosomes separate and move to opposite poles.

Telophse – I

- ✓ One each pair of chromosome is at each pole.
- ✓ Nuclear **possible (membrane)** reappears.
- ✓ Cytokinesis takes place.

ii) Miosis II

Meiosis – II involves the separation of sister chromatids. It occurs through the following stages.

Prophase – II

- ✓ Chromosomes condense
- ✓ Nuclear envelope (membrane) disappears.

Metaphase – II

- ✓ New spindle fibers are formed and sister chromatids line up on it along the cells mid plane.

Anaphase – II

- ✓ Sister chromatids separate and chromosomes more to opposite poles.

Telophase – II

- ✓ Nuclear envelope (membrane) reapers and nuclei is formed at opposite poles of each cell.

Cytokinesis

- ✓ Cytoplasmic division occurs and four daughter cells are formed.

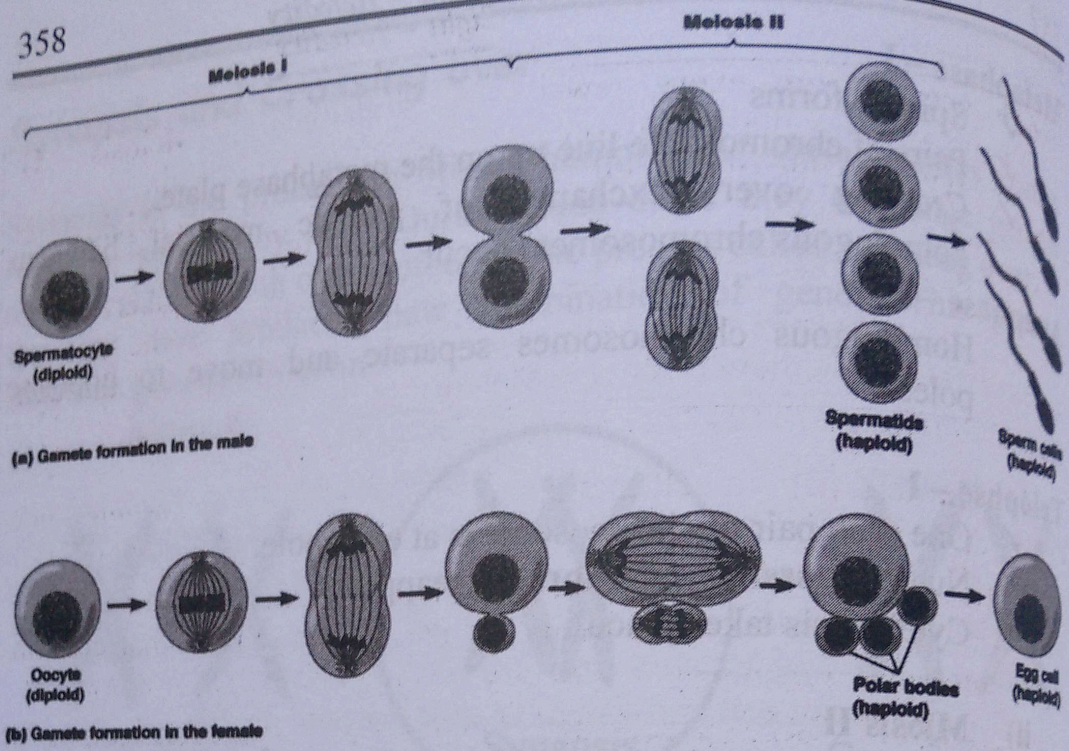


Figure 8.9: Meiosis

Critical thinking
Cells that are produced by mitosis are identical with their parent cell in every aspect. While; cells that are produced by meiosis are not identical with their parent and each other.
Can you explain why and how this occurs?

Comparison of Mitosis and Meiosis

Mitosis	Meiosis
✓ Takes place is somatic cells (body cells). ✓ Daughter cells are similar with the parent cell. ✓ The parent cell and daughter cells have equal number of chromosomes. ✓ Diploid (2n) cells are produced. ✓ Two (2) daughter cells are produced from a single parent cell. ✓ DNA replication always occurs.	✓ Takes place in gonads. ○ The male gonad is testis. ○ The female gonad is ovary. ✓ Daughter cells and parent cells are not similar. ✓ Daughter cells contain half the chromosome number of their parent cells. ✓ Haploid (n) cells are produced. ✓ Four (4) daughter cells are produced from a single cell. ✓ DNA replication occurs only at Meiosis – I. ✓ In girls:

✓ It occurs from every few hours to years depending on the type and age of the cell.	○ First division (Meiosis – I) occurs before birth. ○ Second division (Meiosis – II) occurs monthly starting from the age of puberty. ✓ In boys, starts at the age of puberty and continues through their life.

- **Haploid (n)** – a cell that contains half set of chromosomes, not in a homologous pairs.
  Example: Human sperm and egg cells contain 23 – chromosomes, unlike human body (somatic) cells that contain 46 – chromosomes.
- **Diploid (2n)** – a cell that contains fullest of chromosomes in homologous pairs.
  Example: Human somatic cells contain 46 – chromosomes in 23 – homologous pairs.

Illustrative Example

Item number 15 is based on the following two diagrams of cell division.

Diagram A

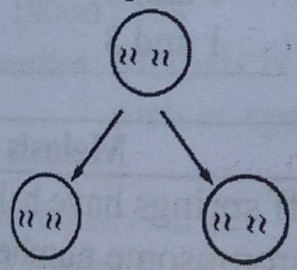
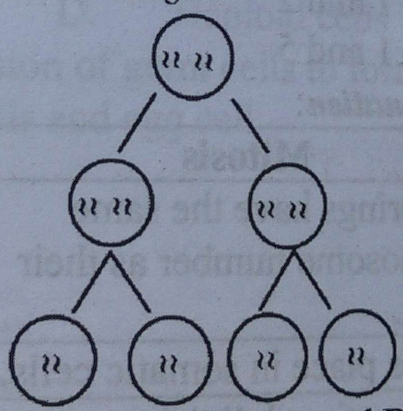


Diagram B



16. Which of the following is true about diagram A and B?
    A. Diagram B is Meiosis cell division.
    B. Diagram A is Meiosis cell division.
    C. Diagram B is the division of somatic cell.
    D. Diagram A is the division of a germ cell.

Explanation:
In mitotic cell division daughter cells have the same chromosome number as a parent cell. While, in meiosis the daughter cells contain half the chromosome number of their parent cell.

Answer: A

17. During meiosis I, in which phase does each chromosome appear in condensed from with two chromatids?
A. Anaphase – I
B. Telophase – I
C. Prophase – I
D. Metaphase – I

Explanation: During prophase – I of meiosis:
- ✓ Chromosomes appear in a condensed form.
- ✓ Nuclear envelope disappears.
- ✓ Homologous chromosomes synapse.

Answer: C

18. This item is based on the following table which shows the comparison between mitosis and meiosis cell division.

No.	Mitosis	Meiosis
1	Offspring will have the same chromosome number of their parent.	Offspring will have half the chromosome number of their patent.
2	It takes place in germ cells.	It takes place in somatic cells.
3	It is a reduction division.	Daughter cells are identical to their parent cells.
4	Two daughter cells are produced.	Four daughter cells are produced.

Which of the above comparison is correctly matched?
A. 1 and 2
B. 1 and 5
C. 1 and 4
D. 1 and 3

Explanation:

Mitosis	Meiosis
Off springs have the same chromosome number as their parent.	Off springs have half the chromosome number of their parent.
It takes place in somatic cells.	It takes place in germ cells.
It is equational division.	It is reduction division.
Two daughter cells are produced.	Four daughter cells are produced.

19. DNA is made up of smaller molecule known as:
A. Gene
B. Nucleotide
C. Phosphate
D. Bases

Explanation: A DNA molecule is made up of smaller molecules called nucleotides. A nucleotide is composed of:
- ✓ Phosphate
- ✓ Deoyribose sugar.
- ✓ Nitrogenous bases (Adenine, thymine, guanine, cytosine)

Answer: B

How do genes control what goes on in the cell and the organism?
A. By initiating the process of cell division.
B. By taking part in the chemical reactions of the cell.
C. By catalyzing chemical reactions that take place in the organism.
D. By specifying the order of amino acids a particular protein.

Explanation: Genes are units of heredity that determine the hereditary characteristics of an organism by specifying the order of amino acids on particular protein.

Answer: D

21. At which stages of meiotic cell division does the spindle form and crossing over takes place?
A. Prophase I
B. Anaphase I
C. Methaphse I
D. Telophase I

Explanation: During metaphase – I
- ✓ Spindle forms
- ✓ Chromosmes line up on the metaphase plate.
- ✓ Crossing over takes place.

Answer: C

22. What is meiosis? It is a division of:
A. Brian cells
B. Blood cells
C. Sex cells
D. Haploid cells

Explanation: Meiosis is a division of germ cells to form gametes such as sperm cells and egg cell.

Answer: C

23. What is mitosis? It is:
A. Formation of sex cells.
B. Copying of chromosomes.
C. Division of somatic cells.
D. Formation of zygote.

Explanation: Mitosis is the division of somatic (body) cells.

Answer: B

24. What is the stage of meiosis where a pair of chromatids moves to opposite poles?
A. Prophase I
B. Anaphase I
C. Anaphse II
D. Metaphase II

Explanation: During anaphase – II
- Sister chromatids separate and chromosomes move to opposite poles.

Answer: C

25. Which of the following happens both in mitosis and meiosis?
A. Two divisions following one another
B. Separation of sister chromatids
C. Separation of homologous chromosomes
D. Reduction of chromosome number by half

Explanation: Both meiosis and mitosis involve the separation of sister chromatids. **Answer: B**

26. What is the amount of DNA in a gamete cell?
A. One fourth of the mother cell
B. Four times that of the mother cell
C. Half of that of the mother cell
D. Twice that of the mother cell

Explanation: Gamete cells like sperm and egg cell contain half the chromosomal number of the parent cells chromosome number. Therefore, the amount of DNA in gamete cell is half of that the mother cell. **Answer: C**

27. Identify the stages in cell cycle, and describe the principal events.

Explanation: Cell cycle is a cyclic series of events in the life of dividing cell. It consists of mainly:

- ✓ **Interphase** – the preparatory stage of cell division, that consists of:
  - Intensive cellular synthesis (G1).
  - DNA replication (s).
  - Mitotic spindles begin to form (G2).
- ✓ **Mitosis (M)** – nuclear division occurs in four phases. Namely, prophase, metaphase, anaphase and telophase.
- ✓ **Cytokinesis (C)** – Cytoplasmic division.

28. The two halves of a replicated chromosome are called:
A. Spindle fibers
B. Chromatids
B Centromeres
D. Centrioles

Explanation:
Sister chromatids are held together by a structure called centromere. Therefore, each of sister chromatids is called **chromatid**. **Answer: C**

29. The correct order of the stages in mitosis is:
A. Metaphase → Anaphase→ Telophase→ Prophase
B. Prophase →Anaphase→ Telophase → Metaphase
C. Metaphase→ Telophase → Anaphase → Prophase
D. Prophase → Metaphase→ Anaphase→ Telophase

Explanation: The correct order of stages in mitosis is:
Prophase → Metaphase → Anaphase→ Telophase

Answer: D

30. What is the number of chromosomes in human fertilized egg?
A. 23 B. 23pair C. 23-46 D. 46 pairs

Explanation: In humans' sperm cell and egg cell each contains 23 chromosomes. Thus, when they get fertilized they form 46 chromosomes (23 pairs).

Answer: B

31. In which of the stages of mitosis do sister chromatids line up along the equatorial plane of the cell?
A. Prophase B. telophase C. anaphase D. metaphase

Explanation:

Prophase	Chromosomes become short and thick Nuclear membrane disintegrate
Metaphase	Sister chromatids line up on the equator
Anaphase	Centromer split and the spindle fibers and chromatids move to the opposite poles
Telophase	Chromosomes become longer and thin nuclear membrane reappears

Answer: D

32. In mitotic cell division, duplication of chromosomes occurs in :
A. Metaphase C. Interphase
B. Telophase D. Prophase

Explanation: Interphase is the longest phase of cell cycle, where the duplication of chromosomes takes place.

Answer: C

33. During ell division centrioles form:
A. Nucleus C. Cenromeres
B. Spindle fibers D. Nucleotides

Answer: B

34. Rearrange the following phases of the cell cycle in their correct order:

Metaphase	Interphase	Anaphase
Prophase	Cytokinesis	Telophase

Explanation: The correct order of phases in a cell cycle are:
Interphase →Prophase→ Metaphase→Anaphase→Telophase

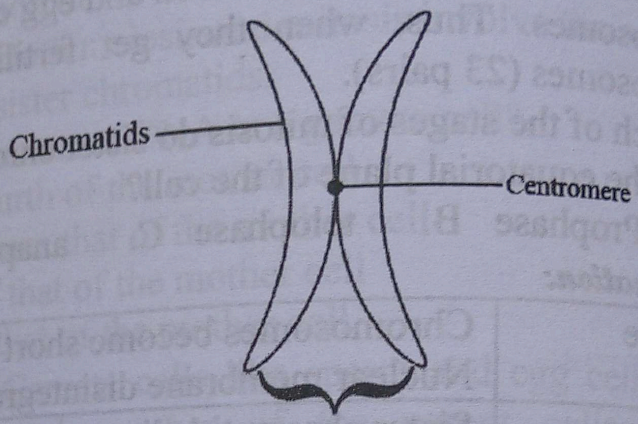
35. Describe the meanings of
- Oogenesis
- Spermatogenesis

Explanation:

- ***Oogenesis*** is the meiotic process of egg formation.
- ***Spermatogenesis*** is the process of sperm formation.

36. Describe the structure of duplicated chromosome including the sister chromatids and centromeres.

Explanation:



37. Explain the significance of mitosis, and describe the process.

Explanation: Mitosis is significant for:

- ✓ Growth of the body.
- ✓ Replacement of damaged and worn out cells.
- ✓ Asexual reproduction.

38. Assume an animal has a chromosome number of 10.
    A. How many chromosomes would it have in a typical body cell such as skin during G1?
    B. How many chromosomes would be present in each daughter cell produced by mitosis?

Explanation:

A. G1 occurs before the duplication of chromosomes. Thus, it has 10 chromosomes.
B. Mitosis results in the production of daughter cells having the same chromosomes number as a parent cell. Thus, the daughter cells produced have 10 chromosomes.

39. Distinguish between haploid and diploid cells.

Explanation:

- ✓ **Haploid** – are cells that have half the chromosome number of a parent cell. They are represented by n. Examples are sperm and egg cells.
- ✓ **Diploid** – are cells that contain full set of chromosome. They are represented by 2n. Examples are all the body cells other than sperm and egg cell.

40. Define
- Homologous chromosomes
- Sister chromatids.

Explanation:

- **Homologous chromosomes** are chromosomes that are similar in morphology and genetic composition. They are pair of identical chromosomes incorporated from both parents.
- **Sister chromatids** – are chromosomes that are copies of one another.

Post Test – II

1. Growth of body tissues is a result of:
   A. Mutation
   B. Meiosis
   C. Metamorphosis
   D. Mitosis
2. If a cell having 8 chromosomes is divided by meiosis, which of the following is **untrue**?
   A. The daughter cells have 4 chromosomes each.
   B. Four daughter cells are produced.
   C. Somatic cells are produced.
   D. Haploid cells are produced.
3. The correct order of stages in meiosis is:
   A. Prophase → Anaphase → Metaphase →Telophase
   B. Anaphase →Telophase→ Metaphase → prophase
   C. Metaphase → Anaphase → Prophase →Telophase
   D. Prophase → Metaphase → Anaphase →Telophase
4. The preparatory stage of cell division is:
   A. Interphase
   B. Cytokinesis
   C. Anaphase
   D. Karyokinesis
5. What is the number of chromosome in a human fertilized egg?
   A. 23 pair B. 23 C. 46 pair D. 92
6. The stage of cell division where nuclear membrane and nucleolus disappears is:
   A. Prophase
   B. Metaphase
   C. Anaphase
   D. Telophase
7. The longest phase of the cell cycle is:
   A. Prophase
   B. Cytokinesis
   C. Interphase
   D. Metaphase
8. The division of nucleus during the process of cell division is called:
   A. Cytokinesis
   B. Karyokinesis
   C. Crossing over
   D. Breaking over

9. Nuclear division occurs during which stage of the cell cycle?
   A. G1 – Phase
   B. S – phase
   C. G2 – phase
   D. M – Phase
10. All the body cells except sex cells are called:
   A. Gametes
   B. Somatic cells
   C. Autosomes
   D. Germ cells
11. All of the following are the importance of mitosis except one, which one is it?
   A. Growth of the body
   B. Sexual reproduction
   C. Replacement of worn out cells
   D. Asexual reproduction

II. Fill in the following blank spaces using an appropriate answer from the list given.

Meiosis	Diploid	Haploid
Replication	Sister chromatids	Meiosis – I
Centromere	Cytokinesis	Meiosis – II
Gametogenesis	Crossing over	

1. _____ Is the process by which a DNA molecule makes it's own copy.
2. A cell that contains full set of chromosomes is _____, while those with half set of chromosomes are called_____.
3. The point of attachment between sister chromatids is known as _____.
4. _____are identical pair of chromosomes formed when a chromosome are copied.
5. _____is known as reduction division.
6. The division of cytoplasm is called_____.
7. _____is the process by which sperm and egg cells are formed.
8. The process by which exchange of genetic materials occurs between homologous chromosomes is called_____.
9. _____involves separation of homologous chromosomes; while, _____ involves the separation of sister chromatids.

8.2 Mendelian Inheritance

Gregor Mendel (1822 – 1884) is the first person who tried explain the mechanisms of in heritance experimentally. Accordingly, he is known as **'the father of genetics.'**

Genetics is a branch of biology which studies that inheritance of traits or characteristics from parents to their offspring. This is also known as **heredity.**

For his experiment of explain the mechanisms of inheritance Gregor Mendel choose a garden pea plant for the following major reasons:

- ✓ It is a self – pollinating plant.
- ✓ It has varieties of contrasting pair of easily identifiable traits.
- ✓ It has short generation time.

In order to show the mechanisms that govern the inheritance of traits by using a garden pea plant; Mendel followed the following major steps:

Step 1: Identification of Contrasting pair of traits

Mendel identified seven contrasting pair of traits of garden pea plant. These are:

Traits	Contrasting pair of traits	
Stem length	Tall	Short
Floral arrangement	Axial	Terminal
Pod color	Green	Yellow
Pod shape	Inflated	Pinched
Seed color	Yellow	Green
Seed shape	Round	Wrinkled
Flower color	Red	White

Do you remember:

- ✓ **Pollination** – is the transfer of pollen grain from another to stigma.
- ✓ **Self – pollination** – a pollination that occurs between the same flower or the same plant having both male and female reproductive organs.

- ✓ **Cross – pollination** – is a pollination that occurs between separate plants. This requires agents that transfer pollen grain from another to stigma. These are wind or animals.
- ✓ **Mendel's artificial** crosspollination was done purposely by transferring pollen grain between garden pea plant having contrasting pair of traits.

Step 2: Self – Pollination

Mendel allowed a garden pea that with each trait (remember table above in which 14 traits identified) to self pollinate for successive generations. This allowed him to check the purity of each trait called **true breed**. Mendel calls each plant with a pure trait **parental generation** (P1).

Example: Tall ×Tall
↓
Tall ×Tall
↓
Tall (pure tall) ⇒ P1 generation (true breed)

Step 3: Artificial Cross – pollination

Mendel conducted an artificial cross pollination between each with contrasting pair of traits. Accordingly, the pea plants produced from this cross are called **first filial (F1) generation**.

Traits	P1 × P1	F1
Stem length	Pure tall × pure short	All tall
Floral arrangement	Pure terminal × pure axial	All axial
Pod color	Pure yellow × pure green	All green
Pod shape	Pure inflated × pure pinched	All inflated
Seed shape	Pure round × pure wrinkled	All round
Seed color	Pure red × pure green	All yellow
Flower color	Pure red × pure white	All white

Step 4: F2 generation

Mendel allowed F1 plants produced to self pollinate and the result was amazingly different. The hidden or masked traits on the F1 generation reappeared in the second filial generation (F2).

$F_1 \times F_1$

↓

F_2

We all know that even though pea plants produced in the F1 generation showed only one trait from the pairing between contrasting pair of traits they contain hidden traits. Thus, they **hybrids** (contain different traits).

Protend: P1 ⇒ Tall × Short

F1 ⇒ Hybrid tall × Hybrid tall

F2 ⇒ 75% tall: 25% short

Trait	F1 ×F1	F2
Stem length	Tall ×Tall	3 tall: 1 short
Floral arrangement	Axial ×Axial	3 axial: 1 terminal
Pod color	Green ×Green	3 green: 1 yellow
Pod shape	Inflated ×Inflated	3 inflated: 1 pinched
Seed shape	Round ×Round	3 round: 1 wrinkled
Seed color	Yellow ×Yellow	3 yellow: 1 green
Flower color	Red ×Red	3 red: 1 white

Conclusion

Mendels Conclusions

On the basis of works on garden pea plant and experimental result, Mendel drew the following conclusions:

1. Since the original parents were pure breed; the tall varieties must have possessed two tall factor and the short variety has two short factors.
2. The F1 generation possessed one factor from each parent which were carried by gametes.
3. The factors in the F1 generation do not blend but retain their individuality.
4. The tall factor is dominant to the short factor which is recessive.

The Mechanisms of Inheritance

As explained in the earlier portion genes located on a chromosome are responsible for determining hereditary characteristics of an organism. A gene consists of alternative forms of traits called **alleles**. Each allele is inherited from both parents and appears in pairs as gene.

For instance, weither you are right handed or left – handed, it is determined by an allele that appears on a gene. This shows that we inherit two possible alleles for a particular feature. One allele from each parent. For example, allele in gene can decide whether:

- ✓ Your earlobes are attached or dangly.
- ✓ Your thumb is straight or curved.
- ✓ You are right handed or left handed.
- ✓ You can roll your tongue or not.
- ✓ You have dimple or not.

Dominant and Recessive Alleles (Genes)

When two alleles, pair on a gene, one trait (allele) dominates or masks the effect of the other. Accordingly,

- ✓ The trait (allele) which expresses itself is called **dominant**.
- ✓ The trait (allele) which is masked is called **recessive**.

For example, in Mendel's experiment tall garden pea plant is dominant over short garden pea plant.

Pure tall × pure short — P1

↓

Tall — F1

- Dominant traits (alleles) are expressed in **capital letter**.
  Example: Tall garden pea plant in Mendel's experiment is dominant and is expressed as 'T'.
- Recessive traits (alleles) are expressed in **small letter**.
  Example: In Mendel's experiment short pea plant is recessive and is expressed as 't'.

Homozygous and Heterozygous gene

- **Homozygous gene** is a gene that contains the same alleles.
  - ✓ It can be:

- **Homozygous dominant** – a gene in which both alleles of the gene are dominant and similar.
  Example: Pure tall pea plant (TT).
- **Homozygous recessive** – a gene in which both alleles of a gene are recessive.
  Example: Pure short pea plant (tt)

• **Heterozygous gene** is a gene in which both alleles are separate.
Example: After pairing pure tall and pure short pea plants in Mendel's experiment, the F1 generation has both tall and short alleles. Thus, it is expressed as 'Tt'.

Genotype and Phenotype

Genotype is the internal (allelic) composition of a gene.

- ✓ It is the allelic composition of an individual for a given genetic trait.

Phenotype is the physical appearance of an individual relating to a given genetic trait.

For example: In Mendel's experiment in the F1 generation the cross between pure tall (TT) and pure short pea (tt) pea plants have produced Tall pea plant. However, tallness is dominant and expressed in the F1 generation.

Tall ⇒ phenotype
Tt ⇒ genotype

Illustrative Example

41. When Mendel crossed plants pure for dominant character and plants pure for recessive character, all the F1 offspring's showed the dominant character. When he crossed the F1 offspring the recessive character appeared in the F2. How could this happen? Because:
    A. The dominant and recessive characters mixed.
    B. The dominant character lost its effect.
    C. The recessive alleles appeared together.
    D. One dominant and one recessive character appeared together.

    Explanation: Recessive alleles are only expressed when they appear together. **Answer: C**

42. What do we call the physical appearance of an organism relating to its genetic makeup?
    A. Alternatives
    B. Phenotypes
    C. Genotypes
    D. Alleles

Explanation:

- ✓ **Genotype** – is the allelic composition of a gene.
- ✓ **Phenotype** – is the external (physical) appearance of an organism.

Example: In Mendels experiment, when the allele for tall and short appear together the external (physical) appearance was tall.

Answer: B

43. Which of the following alleles is expressed phenotypically only when the two copies are found together.
    A. Heterozygous
    B. Hybrid
    C. Recessive
    D. Dominant

Explanation: Recessive alleles are expressed only when the two copies appear in homologous pair.

Answer: C

44. Which of the following biological discoveries is contributed by Gregor Mendel?
    A. The modern cell theory.
    B. The inheritance of characteristics.
    C. The double helix structure of DNA.
    D. The first optical microscope.

Explanation: Gregor Mendel is the father of genetics who explained the mechanisms of inheritance.

Answer: B

45. Distinguish between gene and allele.

Explanation:

- Gene is a unit of heredity that contains two forms of alternative traits (alleles).
- Allele is a an alternative form of gene.

46. What are the components of nucleotide?

Explanation: nucleotide is composed of

- Deoxy ribose sugar.
- Phosphate unit.
- Nitrogenoeous bases.

Test Cross

If an individual has the dominant phenotype, it is not possible to always determine the genotype of this individual by looking at its phenotype. This individual may have homozygous dominant or heterozygous genotype. This can result in the some phenotype.

So, it is better to use test cross to determine organisms' genotype. In a test cross, the individual with unknown genotype is crossed with a homozygous recessive individual.

Punnet Square

Punnet square is a check board that helps to show the possible results of a cross between a given genotype. It is devised by an English genetic Sir. **Reginalt Punnet**.

In punnet square:

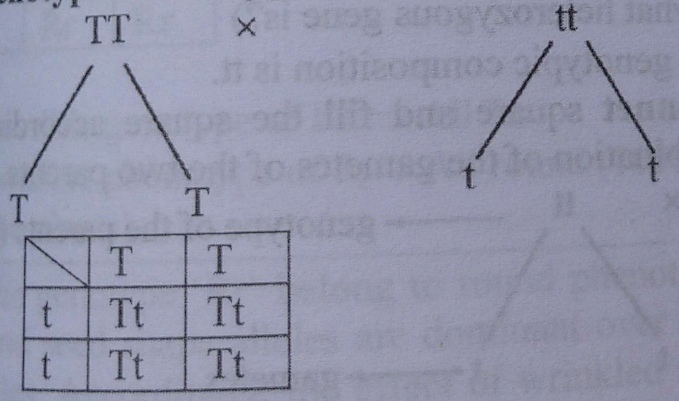
- ✓ the gametes from one parent are listed across at the top.
- ✓ gametes from the other parent are listed along the left side.
- ✓ the squares are filled with the resulting possible genotypes of the offspring's.

Illustrative Example

17. The cross between homozygous tall garden pea plant and homozygous short garden pea plant, according to Mendel's experiment.

Explanation:

Phenotype: Homozygous × short Genotype TT ×tt



	T	T
t	Tt	Tt
t	Tt	Tt

F1 genotype: All Tt

F1 phenotype: 100% Tall

∴ All the F1 generation pea plants had a genotype of Tt. However; because tallness is dominant over shortness, all the pea plants has a phenotype of tall.

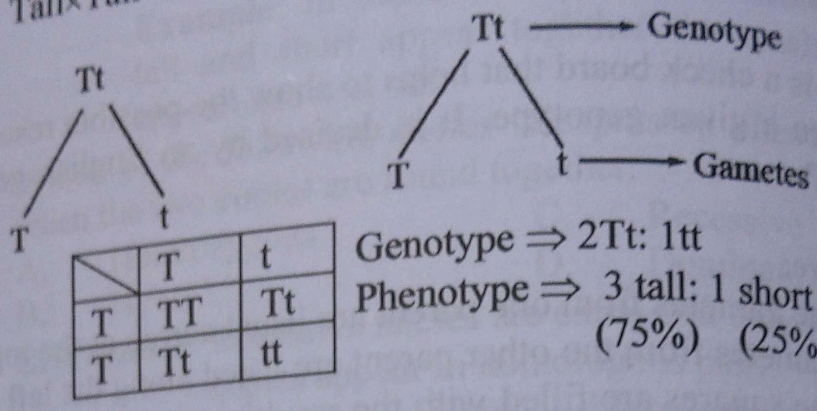
48. The cross between F1 pea plants from above example.

Explanation:

F1 ×F1

Tall×Tall→ Phenotype

Tt×Tt→Genotype



	T	t
T	TT	Tt
T	Tt	tt

Genotype ⇒ 2Tt: 1tt

Phenotype ⇒ 3 tall: 1 short

(75%) (25%)

∴ The genotypes TT and Tt belong to tall phenotype and tt belong to short phenotype.

49. According to Mendel's experiment find the cross between:

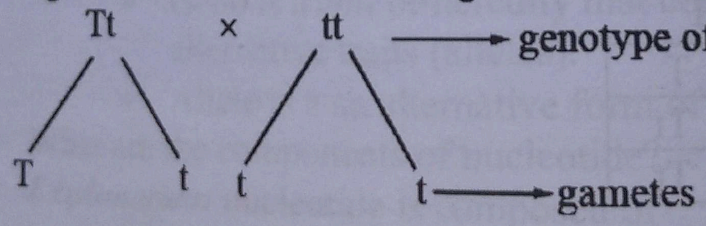
Explanation:

i) Heterozygous tall and short.

Step 1: Find the genotype of both parents.

- ✓ **Heterozygous tall**→ its genotypic composition is Tt, (do you remember what heterozygous gene is?)
- ✓ **Short** → its genotypic composition is tt.

Step 2: Draw a **punnet** square and fill the square according to the genetic combination of the gametes of the two parents.



	T	t
t	Tt	tt
t	Tt	tt

Step 3: List out the possible genotype and phenotypic ration of the F generation produced as a result of the cross between the two parents.

- ✓ Possible Genotypic ratio of F1: 1Tt: 1tt
- ✓ Possible phenotype of F1: 50% Tall: 50% short

ii) Homozygous round and Wrinkled

As seen on Mendel's experiment of garden pea, it has two types of seed shapes. These are; round seed shape (dominant) and wrinkled seed shape (recessive).

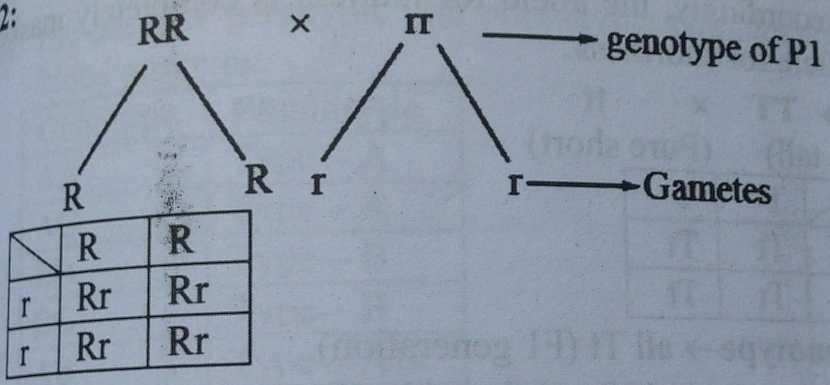
Let us see this example as per the above example.

Step 1:

- ✓ **Homozygous round**→it's genetic composition is 'RR'. (Assume 'R' for round allele.)
- ✓ **Wrinkled**→ its genetic composition is 'rr'. (Assuming 'r' for wrinkled allele.)

Can you explain the reason why we have used 'R' for round seed shape and 'r' for wrinkled seed shape?

Step 2:



	R	R
r	Rr	Rr
r	Rr	Rr

Step 3:

- Possible genotype of F1 ⇒ All Rr
- Possible phenotype of F_1→ All round (100% round seed shaped pea plants).

Note: The genotype 'Rr' belong to round phenotype. This is because round seed shape alleles are dominant over wrinkled seed shape alleles. Accordingly, the effect of wrinkled seed shaped allele is masked by the round seed shape allele.

Quick Check

Try to find out the possible genotype and phenotype of the cross between the following pea plants by referring to Mendel's experiment. (Use punnet square and show your steps).

1) Heterozygous round and Heterozygous round
2) Homozygous red flower and white flower.
3) Heterozygous axial and terminal

Types of Dominance

There are three major types of dominance. These are:

- ✓ Complete dominance
- ✓ Incomplete dominance
- ✓ Co dominance

1) **Complete dominance** – is a condition where one trait (character) completely (fully) masks the effect of the other trait and expresses itself only.
   - ✓ In other words; if two alleles of contrasting traits appear together, the allele for the dominant trait completely masks the effect of the recessive one.

Example: In Mendel's experiment, the cross between pure tall and pure short pea plants results in tall plants in F1 generation. Accordingly, the allele for tallness is completely masked the allele for shortness.

P1 → TT × tt
(pure tall) (Pure short)

	T	T
t	Tt	Tt
t	Tt	Tt

⇒ **Genotype**→ all Tt (F1 generation)
⇒ **Phenotype**→ 100 tall (hybrid tall)
∴ All the pea plants produced in F1 generation were hybrid tall.

ii) **Incomplete dominance**: is a condition when the heterozygote (hybrid) of two alleles has a phenotype that is the intermediate between its two parents.
Example: A cross between Red four o'clock flower and white four o'clock flowers produced pink four o'clock flower, which is an intermediate of both parents.

Red flower × White flower → P1 generation
↓
Pink flower → F1 generation

iii) **Codominance** – is a condition where both alleles in a gene are dominant and so they are both expressed phenotypically.

Examples:

1. When a red bull is crossed with a white cow; resulting claves have roan color. This shows that both the red and white colors are expressed at a time. Some parts of the clave has red coat and some pasts has white coats.
2. Human blood type inheritance

As you have learned on your grade 9 lesson, the human blood grouping is called ABO system. This is for the reason that there are three types of alleles responsible for determining human blood types. These are; allels 'A', 'B' and 'O'.

- Alleles 'A' and 'B' are always dominant and allele 'O' is recessive when it appears with alleles 'A' and 'B'.
- When alleles 'A' and 'B' appear together in a genotype, both alleles are expressed phenotypically and blood group 'AB' is formed.

Table: The possible genotypic and phenotypic combinations in human blood grouping.

Genotype	Phenotype
AA	Type – A
AO	Type – A
BB	Type – B
BO	Type – B
AB	Type – AB
OO	Type – O

Remember: There are one of the four blood types that a person can have. These are:

- ✓ Type – A
- ✓ Type – B
- ✓ Type – AB
- ✓ Type – O

Illustrative Example

50. If a man having blood type A (AO) married a woman with blood type B (BO), what is the possible blood type of their children?

Explanation:

	Father	Mother
Parent phenotype	Type A	Type B
Parent genotype	AO	BO

	B	O
A	AB	AO
O	BO	OO

✓ F1 genotype= 1AB: 1AO : 1 BO 1: OO
✓ F1 Phenotype= 25% type AB, 25% type A, 25% type B, 25% type O

51. If two parents both having blood type A are married and had a child with blood type O; how can you explain to this parents that they should not be surprised by their child blood type?

Explanation:

	Father	Mother
Parent phenotype	Type A	Type A
Possible parent genotype	AA or AO	AA or AO

The possible crosses between the two parents will be:

i) AA XAA

	A	A
A	AA	AA
A	AA	AA

Phenotype: All AA, Genotype 100% Type A

ii) AO X AA

	A	A
A	AA	AA
O	AO	AO

Phenotype= 1AA: AO, Genotype= 100% Type A

iii) AO X AO

	A	O
A	AA	AO
O	AO	OO

Phenotype= 1AA: 2AO: 1OO
Genotype= 75% Type A: 25% Type O
Therefore the parents indicated can have a child with blood type O.

52. A heterozygous yellow flower (Yy) and homozygous green flower (yy) crossed following Mendel is experiment. Which of the following is true about the offspring?
A. All the off spring will have heterozygous yellow flower.
B. All the off springs will have green flower.
C. 25% of the off spring will have green flower.
D. 50% of the off spring will have green flower.

	Y
y	Yy
y	Yy

53. A biology (Yy) pea outcomes
A. Th
B. Al
C. Ty
D. Al

Explana

y	
y	

54. What is alleles f
A. T
B. T
C.
D.

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Explanation: Heterozygous yellow flower (Yy) × Homozygous green flower (yy)

	Y	y
y	Yy	yy
y	Yy	yy

- 1Yy: 1yy – genotypes
- 50% heterozygous yellow: 50% green flower – phenotypes

Answer: D

53. A biology student did an experiment by crossing a yellow seeded (Yy) pea with a green seeded (yy) pea. Which of the following outcomes was correct through the experiment?
A. Three yellow and one green seeded, phenotypically.
B. All of the outcomes were yellow, genotypically.
C. Two yellow and two green seeded, genotypcially.
D. All of the outcomes were green, phenotypically.

Explanation: Yellow seded (Yy) × green seeded (yy)

	Y	y
y	Yy	yy
y	Yy	yy

Genotype = 2Yy. 2yy
Phenotype = 50% yellow: 50% green

Answer: C

54. What is result of a cross between pea plants with heterozygous alleles for round seeds (Rr) and those with wrinkled seeds (rr)?
A. The first generation produces only wrinkled seeds.
B. The first generation produces only round seeds.
C. 50% of the first generation produces round seeds.
D. 75% of the first generation produces wrinkled seeds.

Explanation:
Heterozygous round seeds (Rr) x wrinkled seed (rr) – Parental generation (P1)

	R	r
r	Rr	rr
r	Rr	rr

- Genotype of F1 ⇒ 2Rr: 2rr
- Phenotype of F1 ⇒ 50% round: 50% wrinkled

55. The main objective of Gregor Mendel's work was to prove how:
A. Cell division works.
B. Inheritance works.
C. Breathing system works.
D. Blood circulation works.

Explanation:
The ultimate objective of Gregor Mendel while conducting an experiment on garden pea plant was to prove, how inheritance (transmission of traits) works.

Answer: B

56. Which of the following is true about gene? A gene is
   A. A hereditary material
   B. Located on the cell membrane
   C. A small section of protoplasm
   D. Made up of fats

Explanation: Gene is a unit of heredity which is a hereditary material.

Answer: A

57. Which allele is expressed only when two alleles are present together?
   A. Heterozygous
   B. Hybrid
   C. Recessive
   D. Dominant

Explanation: Recessive alleles are expresses phenotypically only when both alleles in a gene are recessive.

Answer: C

58. Which one of the following statements is correct about the work of Mendel on pea plants? Mendel
   A. Observed that axial flower was a recessive trait.
   B. Removed both the stigma and pollen from the plant.
   C. Observed that round shape was a recessive trait.
   D. Opened his plants before the pollen matured.

Explanation: According to Mendel's experiment on garden pea plant:
   ✓ Axial floral arrangement is dominant.
   ✓ Round seed shape was dominant.
   ✓ Opened his pea plants after the pollen matured.

Answer: C

Mendelian Inheritance in Human

In human being there are more than 25,000 traits that are expressed by genes. Some of these characteristics are controlled by a single gene; while, others are controlled by several genes. Among the several human traits that are controlled by genes, some of them are discussed as follows:

Trait	Dominant	Recessive
Dimple	Dimple	No dimple
Ear lobes	Dangly	Attached
Thumbs	Straight	Curved
Tongue rolling	Roller	Non – roller
Mindedness	Right handed	Left handed

More about human Inheritance

Even though we mostly inherit normal gens from our parents, sometimes people may inherit genes which may cause health problems or even death from their parents. One of these problems is **albinism**.

Albinism

Albinism is the lack of melanin pigment in the skin, hair and eyes. It is a problem mainly in areas where people having black skin live. This is because their skin has high level of melanin.

Melanin is a black pigmented substance in the skin that gives natural protection from the sun. People with paler skin have less melanin and more likely to suffer from skin damage by excessive radiation of sun.

Albinism appears as a result of hidden recessive alleles. It is characterized by:

- ✓ Paler skin
- ✓ Discolored hair

Albino people have increased risk of developing a skin cancer, if they are exposed to excessive sun light.

The normal allele for melanin pigment is dominant and represented by capital letter 'A' and the allele for albinism is recessive and represented by small letter 'a'.

Illustrative Example

59. Suppose an experiment is conducted through crossing of two round shaped (Rr) pea seeds. Which of the following will be possible outcome?
    A. Three offspring will have phenotypically round shape.
    B. Four offspring will have phenotypically round shape.
    C. Four offspring will have genotypically wrinkle shape.
    D. Two offspring will have genotypically wrinkle shape.

Explanation: Rr × Rr

	R	r
R	RR	rr
r	Rr	rr

Genotype: 1RR: 2Rr: 1rr
Phenotype: 3 round: 1 wrinkled

Answer: A

60. The allele to have dangly earlobe is represented by "G" and it is dominant. The allele to have attached earlobes is represented by "g" when they are inherited by the offspring? If the:
A. Offspring inherit "gg" allele from both of their parents, they will have attached earlobes.
B. Offspring inherit "gg" allele from both of their parents, they will have dangly earlobes.
C. Offspring inherit "gg" allele from both of their parents, they will have dangly earlobes.
D. Offspring inherit "Gg" allele from both of their parents, they will have attached earlobes.

Explanation:
- Dangly earlobe is represented by "G" and it is a dominant allele.
- Attached earlobe is represented by "g" and it is a recessive allele.

Accordingly:
✓ 'gg' allele offspring will have attached earlobe.
✓ 'Gg' and 'GG' off springs will have dangly earlobe.

Answer: A

61. Assume both parents (male and female) have a homozygous gene for the expression of attached earlobes. What is the probability that their off springs will inherit the gene for attached earlobes?
A. 75% B. 50% C. 25% D. 100%

Explanation: Attached earlobes are recessive traits. Thus, if two parents who are homozygous for attached earlobes are paired, their offsprings are 100% attached. (g = attached earlobe G = dangly earlobe)

	g	g
g	gg	gg
g	gg	gg

All gg⇒ Genotype
All attached ⇒ phenotype

62. An allele for dangly earlobe 'D' is dominant over an allele for attached earlobes. A woman with dangly ear lobe is married to a man with attached ear lobe. Which of the following shows the possible genotype of their offspring?
A. DD and dd
B. Dd and dd
C. DD
D. DD and Dd

Explanation:
- Dangly ear lobe (D) ⇒ Dominant

- Attached ear lobe (d) ⇒ Recessive

A woman

- **Phenotype = Dangly**
- **Possible genotype** = DD, Dd

i) DD ×dd ii) Dd×dd

Possible crosses

i) DD ×dd ii) Dd×dd

	D	D
d	Dd	Dd
d	Dd	Dd

	D	d
d	Dd	dd
d	Dd	dd

A man

- Phenotype = attached
- Genotype = dd

∴Possible genotypes of off springs are Dd, dd in both cases.

Answer: B

63. If two parents are carriers for albinism, how can they reduce an albino child?

Explanation:

Parental genotype: Aa X Aa

	A	a
A	AA	Aa
a	Aa	aa

✓ Genotype= 1AA: 2Aa:1aa

✓ Phenotype= 75% Normal :25% albino

64. If an albino man marries a normal heterozygous woman. Is there any possibility that they can have a normal child?

Explanation:

	Father	Mother
Parent phenotype	Albino	Heterozygous normal
possible genotype	aa	Aa

	A	a
a	Aa	aa
a	Aa	aa

✓ Phenotype= 1Aa: 1aa

✓ Genotype = 50% normal carrier and 50% albino

65. Suppose 1600 pea plants are produced from TtXtt cross, how many of them are expected to be tall?

Explanation: Parental genotype Tt X tt

	T	t
t	Tt	tt
t	Tt	tt

Genotype= 1Tt: 1tt
Phenotype= 50% tall and 50% short. Therefore, 50% of 1600 is 800.

66. In above cross, how many of them are expected to be TT?

Explanation: None of them are 'TT'

67. According to Mendel's experiment, which generations are pure for traits?

A. P1 plants
B. F1 Plants
C. F2 Plants
D. P1 & F1 plants

Answer:

68. In a fruit fly, grey body (B) is dominant over black body (b). Two grey flies were mated and produced 158 grey and 49 blacks. The parents were probably:

A. BB X BB
B. Bb X Bb
C. BB X Bb
D. Bb X bb

Explanation: Bb X Bb

	B	b
B	BB	Bb
b	Bb	bb

✓ Genotype = 1BB: 2Bb: 1bb
✓ Phenotype= 3 grey : 1 black
The ratio is approximately 3:1

Answer:

69. What do we call the physical appearance of an organism relating to its genetic makeup?

A. Genotypes
B. Alleles
C. Alternative
D. Phenotype

Explanation:

✓ ***Genotype*** is the internal genetic composition of an organism which is expressed in letters.
✓ ***Phenotype*** is the external or physical appearance of an organism.

Answer:

70. Suppose a man with blood group A (AO) married a woman with blood group B (BB). What will be the possible blood groups of their children?

A. 50% AB, 50% BO
B. 75% AB, 25% BO
C. 50%AB, 50% AO
D. 100% AB

Explanation:

	Father	Mother
Parent phenotype	Type A	Type B
possible genotype	AO	BB

	B	B
A	AB	AB
O	BO	BO

✓ Genotype= 1AB: 1BO
✓ Phenotype= 50% Type AB: 50% Type B

Answer: A

71. Which one of the following is a recessive trait in Mendel's experiment?

A. Green pods
B. Tall stems
C. Inflated pods
D. Terminal flower

Explanation:

Mendel's traits	*Dominant*	*Recessive*
Pod color	Green	Yellow
Pod shape	Inflated	Pinched
Stem length	Tall	Short
Floral position	Axial	Terminal

Answer: D

72. In monohybrid cross made by Mendel on garden pea plant, the F1 generation resembled only the one parent. This shows the principle of:

A. Gamete formation
B. Separation of factors
C. Dominance
D. Pair of factors

Answer: C

73. If a father with dangly ear lobe (Dd) married a woman with attached ear lobe (dd); what will be the phenotypic and genotypic ratio of their children?

Explanation:

	Father	Mother
Parent phenotype	Dangly ear lobe	Attached ear lobe
possible genotype	Dd	dd

	d	d
D	Dd	Dd
d	dd	dd

✓ Genotype: 1Dd: 1dd
✓ Phenotype: 50% dangly: 50% attached

74. The black eye allele is (B) is dominant to the brown eye allele (b). What genotype corresponds to the phenotype of black eyed?
A. BB only
B. Bb only
C. Bb and bb only
D. BB and Bb only

Explanation: Black (B) is dominant over the brown eye (b). Therefore, the following genotypes correspond to the black eye color. These are BB and Bb.

Answer: D

75. Abraham has three sisters and both of his parents have homozygous alleles for straight thumbs. How many of his sisters will inherit the allele for straight thumbs?
A. All of his sisters inherit the allele for straight thumbs.
B. One of his sisters inherit the allele for straight thumbs.
C. Two of his sisters inherit the allele for straight thumbs.
D. None of his sisters inherit the allele for straight thumb.

Explanation: If both the parents have homozygous allele, all their children will have their parents trait. Thus, because both parents have homozygous straight thumb allele, the daughters will have all straight thumb.

Answer: A

Post Test – III

I. Choose the best option from the given alternatives.

1. Which of the following genotypes belong to a homozygous dominant character?
A. Tt B. tt C. TT D. TtTt
2. In Mendel's experiment on garden pea, which of the following is a recessive trait?
A. Tall stem length
B. Green pod color
C. Round seed shape
D. White Flower color
3. Mendel choose garden pea plant for his experiment of the inheritance of traits for all of the following, except:
A. It is a self pollinating plant.
B. It has variety of contrasting traits.
C. It is drought resistant.
D. It has short generation time.

4. During Mendel's experimental design, which step comes first?
   A. Artificial cross – pollination
   B. Self pollination
   C. Identification of contrasting traits.
   D. Cross between F2 generations.

5. Assuming Mendel's experiment; a cross between tall garden pea plant and short garden pea plant has produced only tall pea plants. What is the possible genotypic cross between the two parents?
   A. Tt×tt B. Tt×TtC. tt×tt D. TT ×tt

6. Based on information given on item number '5', which trait is recessive?
   A. Tall B. Short

7. As per the information given on item number '5' what are the possible phenotypes on F2 generation?
   A. 50% tall: 50% short C. 100% short
   B. 100% tall D. 75% tall: 25% short

8. Alternative forms of a gene are called:
   A. Chromosomes D. DNA
   B. Alleles D. RNA

9. The external physical appearance of an individual in relation to it's genetic trait is:
   A. Phenotype C. Allele
   B. Genotype D. Test cross

10. The cross between two tall garden pea plants having a genotype of Tt produces:
    A. 25% tall pea plants C. 25% short pea plants
    B. 50% short pea plants D. 50% tall pea plants

II. **Match statements under column 'A' with their complements under column 'B'.**

Column 'A'	Column 'B'
1. Dominant allele	a. Physical appearance of a gene.
2. Genotype	b. Similar alleles in a gene.
3. Punnet square	c. Expresses itself phenotypically when appears in a homozygous.
4. Phenotype	d. Check board to show the possible results of cross between a given genotype.
5. Homozygous	e. Separate alleles in a gene.
6. Heterozygous	f. Internal genetic composition.
7. Recessive allele	g. Expressed always when appears with any type of gene.

8.3 Heredity and Breeding

As mentioned in the previous sections, heredity is the process by which inheritable characteristics pass from parents to their off springs during reproduction.

What is breeding?

Breeding is the process of genetic modification in plants and animals.

- ✓ It is largely important in agriculture for breeding farm animals and crops.
- ✓ It is one of the most widely practiced traditional aspect long before the development of modern biological knowledge's such as heredity. This is also the case in Ethiopia where farmers produce varieties for animals and crop plants through the process of breeding.

Breeding is important for various reasons, but most specially because it gives features that people need from plants and animals. This benefits may include:

- ✓ Getting plenty of milk or many eggs.
- ✓ Producing animals that make lots of muscles to provide meat.
- ✓ Ability of plants and animals to have drought and disease resistant capability, etc.

Group task
Go to a nearby village and ask the farmers about the mechanisms they use while breeding Cattles and crops.

Selective Breeding

Selective breeding is a technique breeding crop plants or animals with desirable traits or economically important individuals in order to preserve only those organisms with best possible traits.
To carry out a selective breeding a crop or animal should be a true (pure) breed having best possible traits. For doing this you need:

- ✓ to know the history the organism or
- ✓ to carry out a test cross to make sure the organism is true breed or homozygous for each trait.

Do you remember?
What is true (pure) breed?

In selective breeding animals with best characteristics and are true breed are allowed to mate. While,organisms with unwanted or undesirable characteristics are castrated or prevented from mating.

Castration – is the technique of making an organism sterile or infertile.

Cross Breeding

Cross breeding is a technique of breeding organisms good (desirable) traits from two different breeds.

For example: Let us take two groups of cattle in which

- One group of cattle are resistant to drought but less production of milk; and
- The other group of cattle has high milk productivity but less resistant to drought.

Therefore, breeding male and female individuals from both groups result in the production of new cattle which have ability to produce high milk yield and resistant to drought.

By using cross breeding we may not get an organism that has the desired characteristics in F1 generation. Thus, it may require breeding for successive generations in order to get an organism with a desired trait.

Importance of breeding for the society

Breeding is important in the society in different forms. Among these some are:

- ✓ By maintaining specialized breeds local farmers can help themselves and keep their international partnership.
- ✓ High yield of crops from plants.
- ✓ High yield of milk and meat form livestock.
- ✓ By cross breeding two breeds having different qualities, it is possible to get combined benefit.

Illustrative Example

76. Which of the following domestic animals in Ethiopia have more breeding types?
A. Goat B. Sheep C. Cattle D. Chicken

Explanation:
In Ethiopia there are 25 types of cattle, 13 types of sheep, 15 types of goats, four types of camels, five types of chicken, four types of donkey, two types of horse and mule. **Answer: C**

77. Which of the following is common feature of cross and selective breeding?
A. Both involve combination of traits.
B. Both improve production yield and quality.
C. Both are used to bread for a particular trait.
D. Both are effective for breeding close relatives.

Explanation:
Both selective breeding and cross breeding are used in improvement of yield and quality of products from animals or crop plants. **Answer: B**

78. The mating between a Borena and a Holstein Friesian cow produce a fertile offspring. This indicates that:
A. They are different species to produce offspring.
B. They look alike morphologically.
C. They are the same species.
D. Their offspring would not be reproduced.

Explanation: Only organisms of the same species can produce a fertile offspring. Thus, the reason why Borena and Holstein Friesian cow produce a fertile offspring is because they belong to the same species. **Answer: C**

79. Which of the following is performed in selective breeding?
Organisms with:
A. Strong and weak characteristics from the different breeds are allowed to mate to get the desired trait.
B. Strong characteristics from the some breed are allowed to mate to get the desired trait.
C. Strong characteristics from different breeds are allowed to mate to produce the desired feature.
D. Strong and weak characteristics from the same breed are allowed to mate to get the desired feature. **Answer: B**

80. Which of the following is true about animal and plant breeding?
A. It reduces genetic diversity.
B. It causes wastage of resources.
C. It provides new and useful genes.
D. It can be done by anybody.

Explanation: Breeding is important, because it:
- ✓ Increases genetic diversity.
- ✓ Reduces wastage of resources.
- ✓ Provides new and useful genes.
- ✓ Done by someone who has knowledge about breeding and the history of the breeds.

Answer: C

81. The type breeding used to improve a breed of organisms through combining good traits from two different breeds is known as:
A. In – breeding
B. Out – breeding
C. Selective breeding
D. Cross – breeding

Explanation: Cross breeding is a technique of breeding organisms with good (desirable) traits from two different breeds.

Answer: D

82. Breeding of animals or plants has a paramount importance to the society of Ethiopia since it.
A. Increase the genetic resources in the country.
B. Decreases the productivity of the animals.
C. Gives low amount of the product we need.
D. Enables society to generate low income.

Explanation: Breeding has several relevance to the society in Ethiopia in various ways:
- ✓ Increase genetic diversity and resources in the country.
- ✓ Increases productivity of animals & plants.
- ✓ Enables the society to generate high income.

Answer: A

83. Which of the following statements is considered as the importance of breeding for a society? It is used to:
A. Reduce genetic diversity.
B. Avoid new genes provision.
C. Reduce the use of natural resources.
D. Produce sufficient food.

Answer: D

84. The type of breeding done in order to get a particular trait is:
A. Multi breeding
B. Cross breeding
C. True breeding
D. Selective breeding

Explanation:

- **Cross breeding** – is a technique of breeding organisms with desirable traits from two different breeds.
- **True breeding** – a genetic strain of an organism in which all individuals have homozygous at the loci under consideration.
- **Selective breeding** – is a technique of breeding organism with desirable trait or economically important individuals in order to preserve only those organisms with best possible traits. It helps to get a particular trait.

Answer: D

85. The process of combining good traits from different breed is:
A. Selective breeding
B. Cross breeding
C. Association breeding
D. Test cross

Answer: B

86. Distinguish between cross breeding and selective breeding.

Explanation:

Selective breeding	Cross breeding
- Is a technique of breeding organisms with desirable traits - It helps to preserve organisms with bets possible traits - Each organism to breed should be true(pure) breed for each trait.	- It I a technique of breeding organisms with desirable traits from two different breeds - Each organism involved is not tested for the purity of each trait to breed.

Review Exercise on Unit – 8

I. Choose the best option from the given alternatives.

1. Of the following nitrogenous bases which one is contained in a DNA molecule?
A. Adenine
B. Thymine
C. Guanine
D. Uracil

2. All of the following are components of nucleotide except:
A. Ribose sugar
B. Deoxyribose sugar
C. Guanine – Cytosine
D. Nitrogenous bases

3. The correct pairing of nitrogenous bases in DNA molecule is
A. Adenine – Cytosine
B. Guanine – Adenine
C. Guanine – Cytosine
D. Thymine–Cytosine

4. DNA differs from RNA in all of the following except one.
   A. It is double stranded.
   B. It mates its own copy.
   C. It contains deoxyribose sugar.
   D. It a nucleic acid.

5. The process of cell division in which daughter cells have the same number of chromosome as their parent cell is:
   A. Mutation
   B. Mitosis
   C. Meiosis
   D. Reproduction

6. A cell with haploid number of chromosome is:
   A. Body cell
   B. Somatic cell
   C. Gamete
   D. Neuron

7. A DNA molecule has 1400 bases and among these, 300 of the bases are Adenine. Accordingly, which of the following shows the correct base number?
   A. Thymine = 400
   B. Guanine = 300
   C. Cytosine = 400
   D. Thymine = 350

8. If the sequence of bases on one strand of nucleotides in a DNA molecule is ATCGTACGT, what will be the sequence of bases on the corresponding chain of nucleotide?
   A. TAGCATGCA
   B. ATCGTACGT
   C. TGCATGCTA
   D. ACGTACGAT

9. In a DNA molecule which of the following is always true?
   A. Adenine = Thymine + Cytosine
   B. Guanine + Cytosine = Adenine + Thymine
   C. Guanine = Adenine
   D. Adenine + Cytosine = Thymine + cytosine

10. Which of the following cell is produced by meiosis?
   A. Nerve cell
   B. Sperm cell
   C. Liver oil
   D. White blood cell

11. The longest phase of the cell cycle is:
   A. Interphase
   B. M – phase
   C. Cytokinesis
   D. Karyokinesis

12. The stage of Mitosis where nucleus and nuclear membrane reappears is:
   A. Prophase
   B. Metaphase
   C. Anaphase
   D. Telophase

13. From the following stages of Mitosis, the correct sequential order is indicated in:
    1. Anaphase
    2. Prophase
    3. Metaphase
    4. Telophase

    A. $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$
    B. $2 \rightarrow 3 \rightarrow 1 \rightarrow 4$
    C. $3 \rightarrow 1 \rightarrow 4 \rightarrow 2$
    D. $4 \rightarrow 3 \rightarrow 2 \rightarrow 1$

14. What occurs during meiosis – I?
    A. Separation of sister chromatids.
    B. Separation of nucleoplasm.
    C. Separation of daughter cells.
    D. Separation of homologous chromosomes.

15. Which of the following sex chromosome combination is correct?
    A. XY chromosome combination results in female.
    B. XX chromosome combination results in male.
    C. XY – chromosome combination results in male.
    D. XXY – chromosome combination results in female.

16. If a cell having 20 chromosomes is divided mitotically, the daughter cells will have how many chromosomes?
    A. 40 B. 20 C. 10 D. 30

17. Which of the following is NOT correct concerning meiosis division?
    A. Haploid daughter cells are formed.
    B. Two daughter cells are formed.
    C. Daughter cells are not identical to the parent.
    D. Gametes are formed.

18. The pair of identical chromosomes inherited from both parents are known as:
    A. Sister chromatids
    B. Identical chromosomes
    C. Daughter chromosomes
    D. Homologous chromosomes

19. The exchange of genetic materials between homologues chromosomes during meiosis is:
    A. Sexual intercourse
    B. Crossing over
    C. Asexual reproduction
    D. Duplication

20. The number of chromosomes in human ovum is:
    A. 46 B. 92 C. 23 D. 22

21. Which of the following trait is NOT inheritable?
    A. Blood type
    B. Scar of the face
    C. Eye color
    D. Skin odour

22. In Mendel's experiment on garden pea plant; the cross between pure tall pea plant and pure short pea plant resulted in tall pea plants on F1 generation. This is an example of:
A. Incomplete dominance C. Complete dominance
B. Concomitance D. Over dominance

23. If two pea plants having homozygous wrinkled and heterozygous round are paired, what conclusions you arrive at?
A. All the F1 plants are round.
B. 25% of the F1 plants are wrinkled.
C. 50% of the F1 plants are wrinkled.
D. 75% of the F1 plants are wrinkled.

24. If a heterozygous dimple man and a woman with no dimple had a child, what will be the possibility of their child to be dimple?
A. 25% B. 50% C. 75% D. 100%

25. Modern genetic is based upon the work of:
A. Charles Darwin C. Gregor Mendel
B. Edward Jenner D. Louis Pastear

26. Growth of body tissues is the result of:
A. Mitosis C. Meiosis
B. Mutation D. Metamorphosis

27. A single cell of a test is divides to produce:
A. A single sperm C. 4 sperms
B. 2 sperm D. 6 sperms

28. A black eye allele (B) is dominant to the brown eye allele (b). What genotypes correspond to phenotypes of black eyes?
A. BB only C. BB and bb only
B. Bb only D. BB and Bb only

29. The nucleotides is a DNA molecule differ in the kind of:
A. Carbon B. Base C. Acid D. Sugar

30. In the experiment of G. Mendel on garden pea plant which of the following are alternate (consulting) traits?
A. Tall × terminal C. Round × Terminal
B. Round × Green D. Tall × Short

31. When round seeds and wrinkled seed of pea plants are crossed, all the hybrid plants produced are round seeds. This shows that wrinkled alleles are:
A. Recessive C. Heterozygous
B. Dominant D. Hybrid

32. If a heterozygous brown eyed man named a blue eyed woman, the genetic makeup of their children will be: (Hint: Brown eye is dominant and blue eye is recessive.)
A. 75% brown and 25% blue eyed.
B. 100% blue eyed.
C. 50% brown and 50% blue eyed.
D. 100% brown eyed.

33. For a given trait, the two genes of an allelic pair are alike. An individual possessing this gene is said to be:
A. Recessive trait C. Heterozygous for that trait
B. Hybrid for a trait D. Homozygous for that trait.

34. A branch of biology which studies about the mechanism of inheritance of traits i:
A. Microbiology C. Reproduction
B. Evolution D. Heredity

35. If a percentage composition of Guanine (G) base in a DNA molecule is 30%, what will be the composition of Adenine (A) base?
A. 30% B. 20% C. 40% D. 60%

36. What do we call 'Mendelian factors' now a days?
A. Genotype C. Gene
B. Chromosome D. Phenotype

37. Which of the following scientists are accredited for the discovery of an amazing DNA molecule?
A. Louis Pasteur and Robert Koch
B. Alexander Fleming and Gregor Mendel.
C. James Watson and Francis Crick
D. Aristotle and Charles Darwin

38. During his experiment on garden pea plant, Mendel made all the pea plants with a particular trait to self pollinate for successive generation. This is because:
A. To check whether the plant can breed or not.
B. To know how self pollination takes place.
C. To check the pority of each trait.
D. To increase the productivity of pea plant.

39. Why do we call meiosis 'a reduction division'?
A. It involves separation of homologous chromosomes.
B. It occurs in small organisms.

C. It produces daughter cells with half a chromosome number of their parent.
D. It takes place in gonads.

40. If a man with blood group O (OO) marries a woman with blood group A (AA), which of the following can be the phenotype of their child?
A. Type – A
B. Type – B
C. Type – O
D. Type – AB

41. Which of the following alleles in human blood type inheritance results in codominance?
A. Allele A and Allele O
B. Allele A and Allele B
C. Allele B and Allele O
D. Allele B and Allele AB

42. An albino man married a normal carrier woman for albinism trait. What is their chance of having an albino child?
A. 25% B. 75% C. 50% D. 100%

43. If two parents whoheterozygous for right handedness trait are married, what is the probability of their children to be left handed? (Right handedness is dominant)
A. $\frac{1}{8}$ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. $\frac{1}{3}$

44. A genetic disorder which is characterized by Paler skin and discolored hair is:
A. Hemophilia
B. Baldness
C. Melanin
D. Albinism

45. A condition where one character masks the effect of the other trait is called:
A. Codominance
B. Complete dominance
C. Incomplete dominance
D. Over dominance

46. The alternative form of a gene is called_____.
A. Allele
B. Chromosome
C. Loci
D. DNA

47. The internal genetic composition of an organism which is expressed in letters is:
A. Phenotype
B. Codominance
C. Genotype
D. Recessive

II. Fill in the blank spaces by using an appropriate answer.

48. The tendency of individuals to resemble their parents is termed as _____.
49. The period from the beginning of one cell division to the beginning of the next cell division is termed as _____.
50. DNA for the new sets of chromosomes is synthesized during _____.
51. The stages of mitosis are divided into four. They are _____, _____, and _____.
52. A duplicated chromosome consists of a pair of _____.
53. The point of attachment of sister chromatids is called _____.
54. The division of cytoplasm to yield two daughter cells is called _____.
55. A pair of chromosomes in which both are inherited from each parent are called _____.
56. Somatic cells contain _____ set of chromosomes, while gametes contain _____ set of chromosomes.
57. Human diploid cells contain _____ chromosomes, while sperm and egg cell contain _____ chromosomes.
58. The physical appearance of an individual organism in relation to a given genetic trait is called_____.
59. A gene that contains both alleles which are similar is known as _____, while a gene that contains separate alleles is called _____.
60. _____are alternative forms of a gene.
61. A pair of identical chromosomes which are formed during interphase are known as _____.
62. _____ Is a photograph of chromosomes in dividing cells which are arranged in pairs.
63. In humans, the male inherits _____ chromosome from his father, and _____ chromosome from his mother.
64. The process by which the sperm cells are formed is called_____

Post Test – I

Answer key for Post Test

1.	D	2.	C	3.	B	4.	B	5.	C
6.	D	7.	B	8.	C	9.	B	10.	C

Post Test – II

Part – I
1. D 2. C 3. D 4. A 5. C 6. A
7. C 8. B 9. D 10. B 11. B

Part – II
1. Replication
2. Diploid, haploid
3. Centromere
4. Sister chromatids
5. Meiosis
6. Cytokinesis
7. Gamete genesis
8. Crossing over
9. Meiosis – I, Meiosis – II

Post Test – III

Part – I
1. C 2. D 3. C 4. C 5. D 6. B
7. D 8. B 9. A 10. C

Part – II
1. G 2. F 3. D 4. A 5. B 6. E 7. C

Answers for Review Exercises on Unit – 8

Part I
1. D 2. A 3. C 4. D 5. B
6. C 7. C 8. A 9. D 10. B
11. A 12. D 13. B 14. 15. C
16. B 17. B 18. D 19. B 20. C
21. B 22. C 23. C 24. B 25. C
26. A 27. C 28. D 29. B 30. D
31. A 32. C 33. D 34. D 35. B
36. C 37. C 38. C 39. C 40. A
41. B 42. C 43. C 44. D 45. B
46. A 47. C

Part – II
48. Heredity 49. Cell cycle 50. Interphase
51. Prophase, metaphase, anaphase, telophase
52. Sister chromatids 53. Centromere
54. Cytokinesis 55. Homologous chromosomes
56. Diploid, haploid 57. 46, 23
58. Phenotype 59. Homozygous, heterozygous
60. Allele 61. Sister Chromatids
62. Karyotype 63. Y, X
64. Spermatogenesis

Unit – 9

9. Human Biology and Health

Introduction

Human body is composed of several cells, tissues, organs and organ systems combined and work together in a coordinated manner. Within this chapter we will clearly discuss different systems such as **nervous system, endocrine system, reproductive system** and **homeostasis.**

9.1 Nervous Coordination

The activities of the body work in a coordinated manner, where different parts work as a team. This enables the metabolic process of our body to perform on the right track. The two main coordinating systems of the human body are:

- ✓ Nervous system and
- ✓ Endocrine system

The nervous system and endocrine systems are shortly summarized in the table below.

Table: Comparison of nervous system and endocrine system

Nervous system	Endocrine System
- Uses a nervous coordination system. - Involves means of electro chemical impulse transmission system. - Nerve impulses are transmitted through/by neurons. - Origins of the message are receptors. - Destination of the message are effectors (muscles and glands) - Provides quick responses - Have short term effect	- Uses a chemical coordination system. - It uses hormonal system. - Hormones are transported through the blood. - Origin of message are ductless gland. - Destination of the message are target organs. - Provides slow response. - Have long lasting effect.

Nervous System

Nervous system is a coordinating system in most animals, including vertebrates and invertebrates.
Nervous system is a very complex system, that provides us rapid coordinated responses to the situations that we meet in our daily activities.
Human nervous system has two sub – systems. They are

- ✓ Central nervous system (CNS) and
- ✓ Peripheral nervous system (PNS)

- **The central nervous system** is composed of **brain** and **spiral cord.**
- **The peripheral nervous system** is composed of all neurons (nerve cells) that run between the **CNS and sensory receptors** found on the sense organs mainly.

Neurons

Neurons are the units of nervous system. **They are smaller units through** which nervous impulses are transmitted.
A neuron is a single nerve cell which is extremely specialized for the transformation of electrochemical impulses or messages.

Structure of a neuron

A neuron is thin and elongated cell. It has the following major parts.

i) **Cell body** – is the structure that contain nucleus, mitochondria, and other organelles.
ii) **Dendrite** – is hair (slender) like cytoplasmic extensions from the cell body to receive signals (impulses) from other nerve cells (neurons).
iii) **Axon** – is a long and thin cytoplasmic extension of the cell body. It carries an impulse from cell body through the neuron.

Mylensheath – is a fatty insulating layer that covers the axon membrane.

- It helps the nerve impulse to travel fastly along the axon.

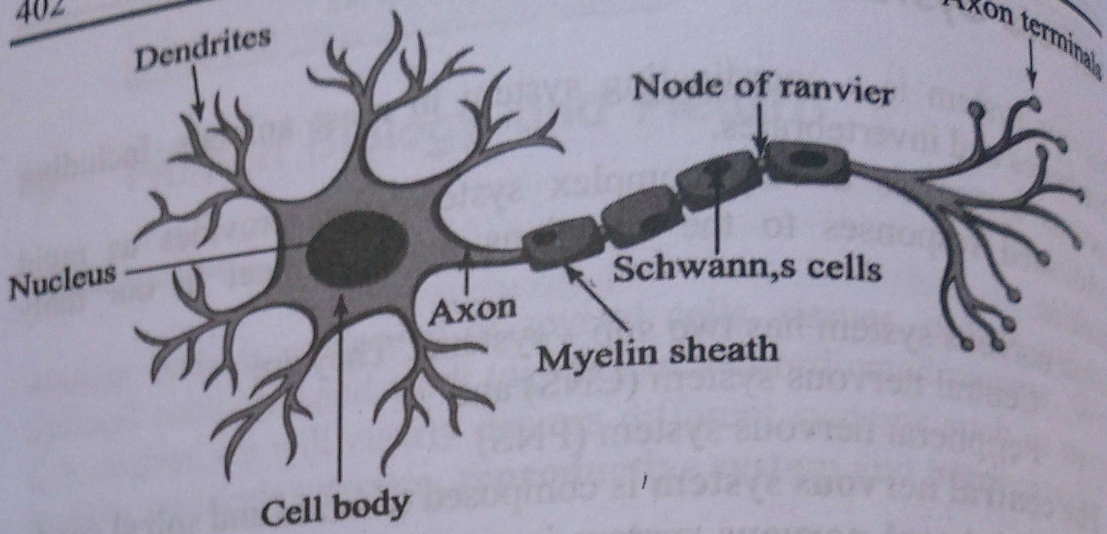


Figure 9.1: A typical neuron

Types of neuron

There are three types of neuron on the basis of their functions. These are:

i) Afferent (sensory) neurons

Afferent neuron:

- ✓ Carries impulse from sensory receptors to the central nervous system (brain and spinal cord).
- ✓ Has very long Dendron to receive an impulse from the receptors.
- ✓ Their cell body is located around sensory receptors.

ii) Efferent (motor) neurons

Efferent neurons

- ✓ Carry impulses from central nervous system to the effectors (muscles and glands).
- ✓ Have a very long axon.
- ✓ Have cell bodies found in the grey mater of the spinal cord.

iii) Relay (association) neuron

Relay neuron a nerve cell that links afferent (sensory) and efferent (motor) neurons with in the central nervous system. They provide transmission of impulse between afferent and efferent nervous within the CNS.

Nerve

Before proceeding into the meaning of nerve, you shall answer the following questions.

> Q. What is the relationship between cell and tissue?
> A. A tissue is a collection of a group of similar or related cells to perform a particular task.

Nerve is a tissue that is composed of a collection/bundle of neurons. Nerve can be classified into 3 types based on the types of neurons they contain. These are:

i) Effector (Motor) nerve

Effector nerve is composed of bundle of efferent neurons. They are responsible for carrying a nerve impulse from the central nervous system to different parts of the body.

ii) Afferent (sensory) nerves

Afferent nerves are nerves that consists of bundles of afferent neurons only. They are responsible for carrying an impulse from sensory receptors to the central nervous system.

iii) Mixed nerves

Mixed nerves are tissues of nerve that consists of the mix of bundles of afferent and efferent neurons.

> **Sensory receptors** are structures that are found on the sense organs of the body. They are used to detect changes (stimulus) that occur inside and outside our body. We have sensory receptors for several types of stimulus; such as heat, light, sound, taste, pain, pressure, touch, etc.

Nerve Impulse

The whole functioning of nervous system was impossible without nerve impulses.
A nerve impulse is a minute electrical event that is the result of charge differences across the membrane of the axon.

Polarized neuron

A polarized neuron a condition where a neuron is not transmitting an impulse. In such a case:

- ✓ The outer membrane of the axon membrane is **positively charged**.
- ✓ The inner portion of the axon membrane is **negatively charged**.

A neuron in such a state is known as **polarized** or **resting neuron**.

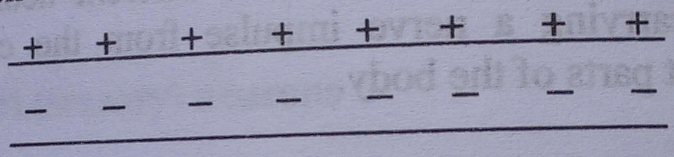


Figure 9.2: polarized neuron

Depolarized neuron

When a neuron starts to transmit an impulse across axon, it gets stimulated. As a result:

- ✓ The positive charge comes inside and the negative charge comes outside the axon membrane.

In such a case the neuron become depolarized. The wave of depolarization axon is called **action potential**.

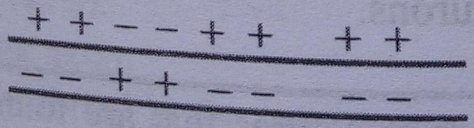


Figure 9.3: Depolarized neuron

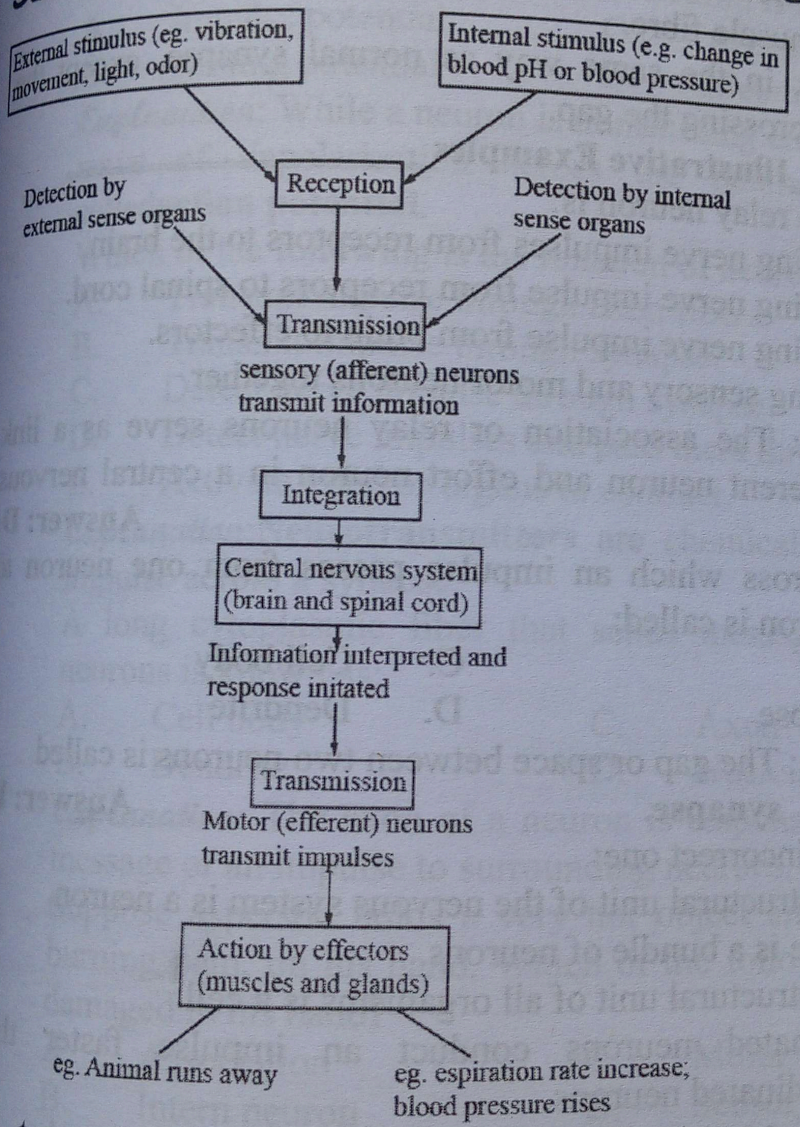
Synapse

Synapse is a gap or a junction between two neurons. This junction can be between:

- ✓ **Axon and dentrite**
- ✓ **A neuron and muscle**

Because there is a gap or a junction between two neurons two neurons do not have an actual connection. Thus, an impulse that is transmitted between two neurons is carried by chemical called **neurotransmitters**. The most commonly known neurotransmitters are **acetylcholine** and **noradrenaline**.

Summary of the Working of Nervous System



The transmission of an impulse across the synapse by neutron transmitters can be affected by poisonous substances. Some of these include:

- ✓ **Butulinus toxin** – affects the first membrane and stops the release of **acetylcholine**. This prevents the transmission of an impulse across the syanpase, which may lead to death.

- ✓ **Strychnine** and **organ phosphorous compounds** – deactivate enzymes on the second membrane and prevent the breakdown of acetylcholine. This causes the nerves to fire continuously and the muscles to contract all the time.
- ✓ **Curare** – stops the second membrane from responding to the neurotransmitters resulting in **paralysis**.

Neuromuscular junctions are junctions between axon terminal of a motor neuron and muscle fibre.

- ✓ They work in the same way as normal synapse, except the chemicals crossing the gap.

Illustrative Examples

1. The function relay neuron is:
   A. Carrying nerve impulses from receptors to the brain.
   B. Carrying nerve impulse from receptors to spinal cord.
   C. Carrying nerve impulse from brain to effectors.
   D. Linking sensory and motor neurons together.

   Explanation: The association or relay neurons serve as a link between afferent neuron and effort neuron in a central nervous system. **Answer: D**

2. The gap across which an impulse passes from one neuron to another neuron is called:
   A. Axon C. Cell body
   B. Synapse D. Dendrite

   Explanation: The gap or space between two neurons is called **synapse**. **Answer: B**

3. Choose the incorrect one:
   A. The structural unit of the nervous system is a neuron.
   B. Nerve is a bundle of neurons.
   C. The structural unit of all organisms is a cell.
   D. Mylinated neurons conduct an impulse faster than unmylinated neurons.

   Explanation: Mylinated neurons conduct or transmit impulse faster than unmylinated neurons. **Answer: D**

4. The dendrites of a neuron can carry an impulse only:
   A. Towards receptors
   B. Away from the cell body
   C. Away from the spinal cord.
   D. Towards the cell body.

Explanation: Dendrites are branched structures that receive impulse and transmit it into a cell body. **Answer: D**

5. Nerve impulses are normally carried toward a neuron cell body by the neurons
   A. Dendrites
   B. Synaptic cleft
   C. Cerebrum
   D. Cerebellum

Explanation: Look at above example. **Answer: A**

6. The wave of depolarization across a neuron is known as:
   A. Impulse potential
   B. Resting potential
   C. Action potential
   D. Membrane potential

Explanation: While a neuron is transmitting an impulse, there is a wave of depolarization across the neuron. This wave is called **action potential**. **Answer: C**

7. Which of the following is the function of neurotransmitters?
   A. Transmit impulse through axon fibers.
   B. Transmit impulse through synapse.
   C. Detect nerve impulse through synapse.
   D. Detect nerve impulses and pass them through dendrites.
   E. Activate effector organs to contract.

Explanation: **Neurotransmitters** are chemicals that transmit an impulse across a synapse. **Answer: B**

8. A long cytoplasmic fiber that sends message to surrounding neurons is known as:
   A. Cell body
   B. Dendrite
   C. Axon
   D. Nucleus

Explanation: The axon of a neuron is responsible for sending a message or an impulse to surrounding neurons. **Answer: C**

9. Suppose a person holds a very hot object and does not feel a burning pain, on his hand. Which of the following are probably damaged in his hand?
   A. Association
   B. Intern neuron
   C. Motor
   D. Sensory

Explanation: Sensory neurons receive an impulse from sensory cells or receptors. Thus, the burning sensation is not felt on the hand, probably due to a damage of sensory neurons. **Answer: D**

10. The nucleus of a neuron is located in the:
    A. Dendrites
    B. Cell body
    C. Synapse
    D. Axon

Explanation: Structures of a neuron such as nucleus and mitochondria are located in the cell body of a neuron.

Answer: B

11. Which of the following structures are effectors?
    A. Motor neurons
    B. Sensory neurons
    C. Muscles
    D. Synapses

Explanation: Effectors are structures that receive an impulse from efferent/motor neurons. These are muscles and glands.

Answer: C

12. Which of the following systems are responsible for the regulation and coordination of body activities?
    A. Nervous and muscular systems
    B. Respiratory and circulatory systems
    C. Nervous and endocrine systems
    D. Nervous and circulatory systems

Explanation: The coordinating systems of the body are nervous system and endocrine system. Both systems work differently towards coordinating body activities.

- ✓ Nervous coordination uses electrochemical impulses.
- ✓ Endocrine system uses hormones/chemical messengers.

Answer: C

13. The cell bodies of motor neurons is located in the:
    A. White matter of the spinal cord.
    B. Grey matter of spinal cord.
    C. Dorsal root of spinal nerves.
    D. Ventral root of spinal nerves.

Explanation:

- The cell bodies of afferent (sensory) neurons is located in the spinal gangalia of the dorsal root of the centerla nervous system.
- The cell bodies of efferent (motor) neurons is located in the grey matter of the spinal cord.

Answer: B

14. The central nervous system consists of:
    A. Brain and spinal cord
    B. Spinal cord and spinal nerves
    C. Anterior and posterior ends of the brain
    D. Cranial and spinal nerves

Explanation: The central nervous system (CNS) consists of the **brain** and **spinal cord**.

Answer: A

15. Which of the following statement is NOT true?
A. A nerve may contain both sensory and motor neuron fibers.
B. Axons covered with mylen sheath transmit impulses faster than naked fibers.
C. Impulses are transmitted in one direction along a neuron.
D. Sensory and motor neurons are always found in separate nerves.

Explanation: A nerve is a bundle of neurons. It may contain:
✓ Only the bundle of motor neurons.
✓ Only the bundle of sensory neurons.
✓ The mix of motor and sensory neurons. **Answer: D**

16. The function of association neurons is:
A. Carrying nerve impulses from receptors to the brain.
B. Carrying nerve impulse from receptors to spinal cord.
C. Carrying nerve impulse from brain to effectors.
D. Linking sensory and motor neurons.

Explanation: The function of association neurons is connecting the sensory and motor neurons. **Answer: D**

17. The gap across which an impulse passes from one neuron to another is
A. Axon
B. Synapse
C. Dendrite
D. Cell body

Explanation: Synapse is a junction or a gap between two neurons. **Answer: B**

18. After information is processed in the central nervous system instructions are sent to the body by specific neurons, which are known as:
A. Afferent neurons
B. Sensory neurons
C. Affecter neurons
D. Efferent neurons

Explanation:
✓ **Afferent neurons** – carry impulse from the sensory receptors to the CNS.
✓ **Efferent neurons** – carry instructions from CNS to effectors
✓ **Effectors** – are glands and muscles of the body carry receive instructions from the CNS.

19. A nucleus of a neuron is located in the:
A. Axon
B. Dendrites
C. Synapse
D. Cell body

Explanation: A nucleus and organelles of a neuron is located in the cell body. **Answer: D**

20. Choose the incorrect statement:
    A. The structural unit of the nervous system is neuron.
    B. The reflex arc is the functional unit of nervous system.
    C. The structural unit of all organisms is cell.
    D. None of the above

Explanation: Reflex arc is a path of an impulse during reflex action. **Answer: B**

21. List out the major parts of the neuron.

Explanation: A neuron has the following major parts:
- ✓ Cell body
- ✓ Axon
- ✓ Dendrite

22. Mention the difference between a neuron and a nerve.

Explanation:
- A neuron is a single nerve cell while a nerve is a bundle of neurons.

23. What is a role of neurotransmitters in nerve impulse transmission?

Explanation: Neuron transmitters are chemicals that carry an impulse along the axon.

24. Explain the meaning of action potential.

Explanation: Action potential is a wave of depolarization across the axon.

Post Test – 1

I. Choose the best answer from the given alternatives.

1. Endocrine system differs from a nervous system because:
   A. It uses electrochemical impulse transmission.
   B. Uses chemical coordination systems.
   C. Messages are transmitted through neurons.
   D. Provides quick response.
2. Which of the following are coordinating systems of the body?
   A. Endocrine system
   B. Nervous system
   C. Nervous system and endocrine system
   D. Nervous system
3. Structure of the neuron that contains nucleus and mitochondria is:
   A. Cell body
   B. Dendrite
   C. Axon
   D. Synapse

Fatty insulting layer of the axon is:
A. Synaptic cleft
B. Mylen sheath
C. Axon membrane
D. Fatty acid

A type of neuron that carries an impulse to the central nervous system is:
A. Efferent neuron
B. Afferent neuron
C. Relay neuron
D. Motor neuron

Afferent and efferent neurons are connected by:
A. Relay neuron
B. Motor neuron
C. Sensory neuron
D. Afferent neuron

Structures found on sense organs and can detect stimulus are:
A. Effectors
B. Receptors
C. Glands
D. Neurons

A neuron at rest is called:
A. Depolarized
B. Polarized
C. Respolarized
D. Deactivated

9. _____is the wave of depolarization across the neuron.
A. Resting potential
B. Action potential
C. Axon potential
D. Nerve potential

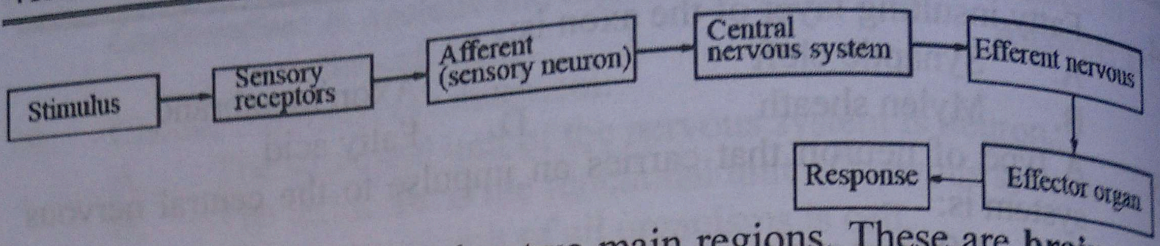
10. A junction between two neurons is:
A. Axon
B. Synapse
C. Dendirte
D. Mylea sheath

11. Chemicals that carry an impulse across the synapse are:
A. Chemoreceptors
B. Chemo transmitters
C. Neurotransmitters
D. Axon

12. Among the toxics that affect the working of neuron transmitters, paralysis may result by:
A. Bustulines toxin
B. Strychinine
C. Curare
D. Organphosphrous compound

The Central Nervous System (CNS)

The central nervous system is a part of a nervous system where the electrical messages that come from the entire body are coordinated. As result our sense organs get an information from CNS to respond to changes in our external and internal environments.
Without the processing or coordination of the CNS, the impulses that are carried from receptors by afferent neurons are of no use at all. The pathway of an impulse during nervous coordination is traced as:



The central nervous system has two main regions. These are **brain** and **spinal cord**.

The Brain

The brain is a delicate organ that is composed of mass of nervous tissues. An adult human brain weighs in average around 1.3 to 1.4 kilogram.

Because the brain is a very delicate organ, it is surrounded and protected by different structures. These are:

- ✓ **Skull or cranium** – is the bone of head in which the brain is enclosed. It is very strong and provides protection from mechanical damage.
- ✓ **Meninges** – is a system of membranes that surrounds and protects the brain.
- ✓ **Cerebro spinal fluid** – are fluids in which the brain suspends. They protect the brain from knocks and blows.

The brain is connected to different parts of the body by means of:

- ✓ **Cranial nerves** – come out of the brain and directly connect the brain with structures in the head and necks. These include eyes, tongue, jaws, etc.
- ✓ **Spinal nerves** – are most of the nerves in our body. They pass through the spinal cord and connect several part of the body to the brain. These body parts include: arms legs, trunk and the rest of the body, excluding parts of neck and head.

The human brain consists of different areas that carryout different functions.

The bulk of the brain is made up of the **grey matter** which consists of cell bodies of neurons and synapses. The inner portion of the brain is the **white matter** which is composed of axons that lead into and out the brain.

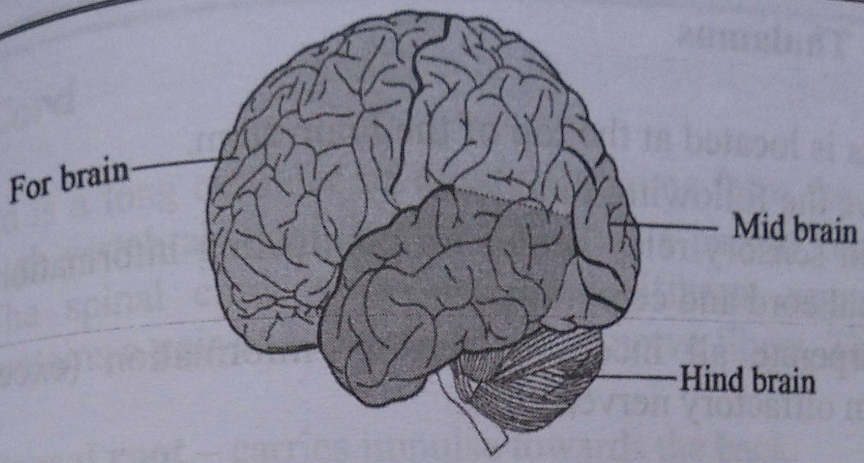


Figure 9.4: The brain

The human brain is sub-divided into three main regions. These are:

- ✓ Forebrain
- ✓ Mid brain
- ✓ Hind brain

1. The Fore brain

The forebrain is the largest part of the brain. It have different regions, but mainly the **cerebral (cerebral hemisphere)**.

The other regions of the fore brain ore also **thalamus** and **hypothalamus**.

i) Cerebrum (Cerebral hemisphere)

Cerebrum are divided into left and right sides called **cerebral hemisphere**.

- ✓ The left cerebral hemisphere is concerned with processing logical and mathematical abilities.
- ✓ The right cerebral hemisphere is concerned with artistic abilities.

The cerebrum is involved in coordinating higher levels of activities such as:

- ✓ Thinking
- ✓ Speech
- ✓ Memory
- ✓ Intelligence

Each side of our body is controlled byopposites sides of the brain.

- ✓ The activities of the left hand side of the body is controlled by the right side of the brain and vice versa.

The left hand side of the brain is good at processing information and controls activities such as speech. The right hand side of the brain is better at special awareness and recognition of faces of people.

ii) Thalamus

- Thalamus is located at the top of the brain stem.
- They have the following functions:
  - ✓ Main sensory relay center for conducting information between spinal cord and cerebrum.
  - ✓ Interperate all incoming sensory information (except) those from olifactory nerves.

iii) Hypothalamus

Located just below thalamus and connected to pituitary gland (you will learn about pituitary gland under endocrine system).
Hypothalamus consists of centers for regulating and controlling various activities. These are:

- ✓ Regulation of body temperature
- ✓ Appetite – feelings of hunger and thirst
- ✓ Certain emotions
- ✓ Fat metabolism
- ✓ Regulation of pituitary gland

2. Mid Brain

Mid brain is found just in between the fore brain and hind brain. It consists of an optic lobe; a center for vision.

3. Hind brain

Hind brain is the lower portion of the brain that consists of the following structures:

- ✓ **Cerebellum** is a reflex center for muscular coordination and refinements of movements. It is directly involved activities such as:
  - Maintenance of body balance and posture.
  - Voluntary movements of the body
- ✓ **Medulla oblongata** – contain vital nerve centers that control heartbeat, respiration, blood pressure, breathing, salivation, swallowing, coughing vomiting.
- ✓ **Pons** – connects various parts of the brain with one another. They also contain respiratory centers and sleep centers.

Spinal Cord

Spinal cord is a long cylinder of nerves that runs from the base of the brain through vertebral canal (back bone). It is a part of center nervous system. The spinal cord is divided into different segments. Each segment contains a pair of roots made out of nerve fibers. The two roots are:

- ✓ **Dorsal root** – carries impulse towards the back.
- ✓ **Ventral root** – carries impulse away from the root.

Spinal cords have the following major functions. They:

- ✓ Create communication between different sections of the body and with the brain.
- ✓ Allow waking (locomotion)
- ✓ A center for a reflex action

A cross section of the spinal cord

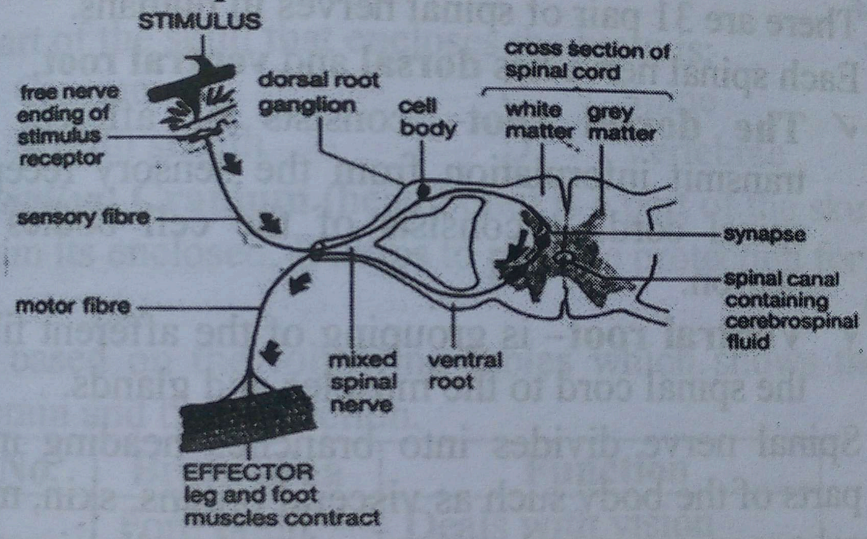


Figure 9.5: spinal cord

The butterfly shaped region of the spinal cord is called **grey matter**, while the surrounding while materials called **white matter**.

- **White matter** – consists of nerve fibers called axons that extend the length of the cord. It allows part of the CNS to communicate.
- **Grey matter** – consist of dorsal and ventral root which consist of cell bodies of neurons.

The Peripheral Nervous System

Peripheral nervous system (PNS) consists of the sensory receptors, the nerves that link the CNS with the effectors (muscles and glands).
The PNS consists of two types of nerves. These are:

- ✓ **Cranial nerves** – are nerves that directly emerge from the brain stem.
  - They connect structures such as head, neck, eyes, tongue, jaws, etc. with the brain.
  - In human body there are 12 pair of cranial nerves.
  - They transmit information regarding the senses of smell, sights, hearing and generally all the sensory receptors from the head region.
- ✓ **Spinal nerves** – are nerves that emerge from spinal cord.
  - There are 31 pair of spinal nerves in humans.
  - Each spinal nerve has **dorsal** and **ventral root**.
    - ✓ **The dorsal root**– consists of afferent fibers that transmit information from the sensory receptors to the spinal cord. It consists of the cell bodies of sensory neuron.
    - ✓ **Ventral root**– is grouping of the afferent fibers leaving the spinal cord to the muscles and glands.
  - Spinal nerve divides into branches heading into different parts of the body such as visceral organs, skin, muscles, etc.

The peripheral nervous system has two main sub – divisions. These are

- ✓ Somatic division
- ✓ Automatic division

i) In somatic division of the PNS:
- ✓ **Afferent (motor) nerves** – transmit information from receptors of CNS.
- ✓ **Efferent (motor) nerves** – transmit information from CNS to skeletal muscles.

ii) In automatic division:
- ✓ **Afferent (sensory) nerves** – transmit information from receptors in internal organs to CNS.
- ✓ **Efferent nerves** – transmit information from CNS to glands and involuntary muscle in organs. These can be sub – divided as:

- **Sympathetic nerves** – generally stimulate activity that results in mobilization of energy (e.g. speeds heart beat)
- **Parasympathetic nerves** – action results in energy conservation or restoration (E.g. Slows hear but speeds digestion)

Illustrative Example

The following item is based on the following list of structures for impulse transmission.

1. Sense organ
2. Central nervous system
3. Muscle
4. Afferent neuron
5. Efferent neuron

25. The correct order of impulse transmission is

A. $1 \rightarrow 4 \rightarrow 2 \rightarrow 5 \rightarrow 3$
B. $1 \rightarrow 3 \rightarrow 2 \rightarrow 4 \rightarrow 5$
C. $1 \rightarrow 5 \rightarrow 2 \rightarrow 4 \rightarrow 3$
D. $1 \rightarrow 4 \rightarrow 3 \rightarrow 5 \rightarrow 2$

Explanation: The correct order of impulse transmission is:

$1 \rightarrow 4 \rightarrow 2 \rightarrow 5 \rightarrow 3$

Answer: A

26. The part of the skull that encloses the brain is:

A. Cranium
B. Myelin sheath
C. Spine
D. Vertebrae

Explanation: **Cranium** (head bone) is a part of the skull in which the brain its enclosed. It helps to provide protection for the brain.

Answer: A

This item is based on the following tables which shows the different areas of the brain and their function.

No.	Brain area	Function
1	Fore brain	Deals with vision
2	Mid brain	Deals with breathing
3	Hind brain	Deals with balancing

27. Which of the following brain area is correctly matched with its function?

A. 1 and 3
B. 1 and 2
C. 3
D. 2

Explanation: The correct match of the brain area and function is indicated as:

Brian area	Function
Fore brain	deals with major activities such as speech, thinking, memory, intelligence, etc
Mid brain	deals with vision
Hind brain	deals with balance, breathing, circulation etc.

Answer: C